

# Eleven Primitives and Three Gates: The Universal Structure of Computational Imaging

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## Abstract

Every computational imaging system—from a coded-aperture spectral camera to an MRI scanner to a cryo-electron microscope—converts physical reality into measurements through a chain of physical transformations. Despite the apparent diversity of these systems across carrier families, length scales, and modalities, here we show that the space of all such forward models has *finite structural complexity*. We prove the **Finite Primitive Basis Theorem**: every imaging forward model in a broad operator class admits an  $\varepsilon$ -approximate representation as a directed acyclic graph over exactly 11 physically typed primitives—a basis that is both sufficient and minimal<sup>1</sup>. This is a discovery, not a design choice: the primitive library saturates at 11 as modality coverage grows past 168 registered systems across 19 categories, with no new primitive required for the most recent 138 additions, proving that the structural complexity of imaging physics is bounded. We further establish the **Triad Decomposition**, a universal diagnostic law governing every failure in computational image recovery: reconstruction error decomposes into exactly three independent root causes—information deficiency, carrier noise, and operator mismatch—mirroring the role of entropy, thermal noise, and free-energy barriers in the thermodynamics of irreversible processes. Hardware validation across twelve modalities spanning all five carrier families—photons, electrons, X-rays, nuclear spins, and acoustic waves—including cryo-electron microscopy and clinical cone-beam CT, confirms that operator mismatch, not information deficiency or noise, is the dominant reconstruction bottleneck under standard operating conditions, recoverable by +0.8 to +10.7 dB ( $\pm 0.3$  dB, 95% bootstrap confidence interval) through forward-model correction alone—often exceeding the gain from upgrading a classical solver to a state-of-the-art deep network. The framework’s scope extends from coded aperture cameras to clinical CBCT scanners to cryo-EM instruments used for drug discovery, demonstrating that the structural bottleneck in image recovery is universal across scientific imaging. These results shift the primary locus of investment in computational imaging from solver design to calibration, and provide a unifying blueprint for decomposing forward models across physical measurement science.

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## 33 Introduction

34 Every imaging instrument—whether it captures spectra, spins, X-rays, or sound waves—  
 35 converts a physical scene into digital measurements through a chain of transformations:  
 36 a wave propagates, interacts with the object, passes through optics, and is recorded by  
 37 a detector. Computational imaging reconstructs the scene by inverting this chain. The  
 38 reconstruction algorithm assumes a mathematical model of the chain (the “forward model”),  
 39 but this model inevitably diverges from reality: optics shift, detectors drift, and calibration  
 40 degrades. When the model is wrong, the reconstruction fails—not because the algorithm is  
 41 weak, but because it is solving the wrong problem.

42 This paper establishes that the space of imaging forward models has a surprisingly  
 43 simple and universal structure, and that exploiting this structure reveals operator mis-  
 44 match as the systematic, cross-modality bottleneck in reconstruction quality. Over the  
 45 past decade, the community has invested enormous effort in improving reconstruction  
 46 algorithms—progressing from compressed sensing<sup>2,3</sup> and plug-and-play priors<sup>4,5</sup> to deep  
 47 unrolling networks<sup>6</sup> and vision transformers<sup>7</sup>—with snapshot compressive imaging emerg-  
 48 ing as a unifying framework for coded aperture systems<sup>8</sup>—yet these algorithms routinely fail  
 49 when deployed on real instruments<sup>9</sup>. Calibration methods exist for specific instruments—  
 50 ESPIRiT<sup>10</sup> for MRI, ePIE<sup>11</sup> for ptychography—but they do not generalise across modal-  
 51 ities, and no systematic framework decomposes reconstruction failures into root causes.

52 Here we establish two complementary foundations within Physics World Models (PWM).  
 53 The **Finite Primitive Basis Theorem** (Theorem 3) proves that every imaging forward  
 54 model in a broad operator class  $\mathcal{C}_{\text{img}}$  admits an  $\varepsilon$ -approximate representation as a typed  
 55 directed acyclic graph (DAG) over exactly 11 canonical primitives—and that this library is  
 56 minimal<sup>1</sup>. The **Triad Decomposition** proves that every reconstruction failure decomposes  
 57 into three gates—information deficiency (**Gate 1**), carrier noise (**Gate 2**), and operator  
 58 mismatch (**Gate 3**)—with a formal condition (Theorem 5) under which **Gate 3** dominates.  
 59 The 11 primitives *enable* the 3-gate diagnosis: the DAG structure localizes mismatch to a  
 60 specific physical operator, making correction actionable.

61 We validate both results across twelve modalities spanning all five carrier families—  
 62 optical photons (CASSI<sup>12</sup>, CACTI<sup>13</sup>, SPC<sup>14</sup>, lensless, compressive holography, fluores-  
 63 cence microscopy), X-ray photons (CT<sup>15</sup>, cone-beam CT), electrons (ptychography<sup>16</sup>, cryo-  
 64 EM), nuclear spins (MRI<sup>17,18</sup>), and acoustic waves (ultrasound)—with hardware validation  
 65 on ten modalities confirming Triad predictions on physical measurements. The validated  
 66 modalities include two Nobel Prize-winning techniques (cryo-EM, 2017 Chemistry; super-  
 67 resolution fluorescence, 2014 Chemistry), three clinically deployed instruments (CT, CBCT,  
 68 MRI), and the first acoustic-carrier validation in computational imaging. A held-out closure  
 69 test on 8 additional modalities confirms basis completeness with saturating growth. The  
 70 proof of Theorem 3 is in Supplementary Note 12; formal primitive semantics and typed  
 71 DAG denotation are developed in<sup>1</sup>.

72 **Why this matters beyond computational imaging.** Any physical measurement system—  
 73 seismographs, telescopes, particle detectors, environmental sensor networks—faces the same  
 74 structural problem: a forward model maps the quantity of interest to observations, and  
 75 model fidelity limits what can be recovered. The finding that eleven primitives span the  
 76 entire space of imaging forward models raises a fundamental question about the grammar  
 77 of physical measurement: if five carrier families, twelve validated modalities, and 168 regis-  
 78 tered systems across 19 categories all compose from the same small set of operators, is this  
 79 a contingent feature of imaging, or does it reflect a deeper constraint on how information  
 80 flows from physical systems to digital observations? The OPERATORGRAPH formalism pro-  
 81 vides a template for decomposing forward models in other domains into typed, composable  
 82 primitives with validated adjoints. The Triad Decomposition—separating information ca-  
 83 pacity, noise, and model fidelity as independent failure modes—applies wherever an inverse  
 84 problem is solved computationally. The practical lesson is domain-general: before investing  
 85 in a better solver, diagnose whether the model is the bottleneck.

## 86 The Finite Primitive Basis

87 A foundational question in computational imaging is whether the diversity of imaging for-  
 88 ward models—from coded aperture cameras to MRI scanners to electron microscopes—can  
 89 be captured by a small, fixed set of primitive operators. We answer affirmatively.

90 **The OperatorGraph representation.** Every imaging forward model is encoded as a  
 91 typed directed acyclic graph (DAG) in which each node wraps a single primitive physical  
 92 operator and edges define the data flow from source to detector. Every primitive implements  
 93 both a `forward()` and an `adjoint()` method, with validated adjoint consistency ensuring  
 94  $\langle H\mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{x}, H^\dagger \mathbf{y} \rangle$  to within numerical precision. Each edge carries tensor shape and  
 95 dtype metadata, enabling static validation before execution. We call this formalism the  
 96 OPERATORGRAPH intermediate representation (IR).

97 **Eleven canonical primitives.** The OPERATORGRAPH IR defines a library of exactly 11  
 98 canonical primitives  $\mathcal{B} = \{P, M, \Pi, F, C, \Sigma, D, S, W, R, \Lambda\}$ :  
 99 The Detect nonlinearity  $\eta$  is restricted to five canonical families (linear-field:  $\eta(x) = gx$ ;  
 100 logarithmic; sigmoid; intensity-square-law:  $\eta(x) = g|x|^2$ ; coherent-field), each with at most  
 101 2 scalar parameters. The Transform primitive  $\Lambda$  applies a pointwise nonlinear function  
 102 within the physics chain (not at the detector) and is restricted to five families: exponential  
 103 attenuation ( $e^{-\alpha x}$ , Beer–Lambert), logarithmic compression, phase wrapping ( $\arg(e^{ix})$ ),  
 104 polynomial ( $\sum a_k x^k$ ,  $d \leq 5$ ), and saturation. Both Detect and Transform are pointwise  
 105 and parameter-limited, preventing either from becoming a universal approximator (see<sup>1</sup> for  
 106 formal definitions and non-universality proofs).

| #  | Primitive  | Notation                          | Physical action                                      |
|----|------------|-----------------------------------|------------------------------------------------------|
| 1  | Propagate  | $P(d, \lambda)$                   | Free-space wave propagation                          |
| 2  | Modulate   | $M(\mathbf{m})$                   | Element-wise multiplication (mask, coil, absorption) |
| 3  | Project    | $\Pi(\theta)$                     | Radon line-integral projection                       |
| 4  | Encode     | $F(\mathbf{k})$                   | Fourier-domain encoding ( $k$ -space)                |
| 5  | Convolve   | $C(\mathbf{h})$                   | Spatial convolution (PSF)                            |
| 6  | Accumulate | $\Sigma$                          | Summation over spectral/temporal axis                |
| 7  | Detect     | $D(g, \eta)$                      | Detector response (5 canonical families)             |
| 8  | Sample     | $S(\Omega)$                       | Sub-sampling on index set $\Omega$                   |
| 9  | Disperse   | $W(\alpha, a)$                    | Wavelength-dependent spatial shift                   |
| 10 | Scatter    | $R(\sigma, \Delta\varepsilon)$    | Direction change and/or energy shift                 |
| 11 | Transform  | $\Lambda(f, \boldsymbol{\theta})$ | Pointwise nonlinear physics operation                |

**Physics-stage mapping.** Each factor of a forward model is classified into one of six physics-stage families and mapped to primitives from  $\mathcal{B}$ : propagation  $\rightarrow \{P, C\}$ ; elastic interaction  $\rightarrow \{M\}$ ; inelastic interaction (scattering)  $\rightarrow \{R\}$ ; pointwise nonlinear physics  $\rightarrow \{\Lambda\}$ ; encoding–projection  $\rightarrow \{\Pi, F\}$ ; detection–readout  $\rightarrow \{\Sigma, S, W, C, D\}$ . This classification formalizes the source–medium–sensor decomposition that structures classical imaging theory<sup>19</sup>.

**Fidelity levels.** Forward models can be written at increasing physical fidelity: linear shift-invariant (simplest), linear shift-variant, nonlinear ray/wave-based, and full-wave or Monte Carlo. All 11 primitives apply uniformly across every fidelity level. Linear stages are composed from the first 10 primitives; nonlinear stages (beam hardening, phase wrapping, saturation) are captured by Transform  $\Lambda$ . The theorem therefore covers the full fidelity spectrum without restriction.

**Five physical carriers.** The library spans five carrier families—photons, electrons, spins, acoustic waves, and particles—with 168 registered modality templates across 19 categories (microscopy, medical imaging, electron microscopy, remote sensing, spectroscopy, and others; see<sup>1</sup>), of which 12 now have full end-to-end correction validation across all five carrier families (Extended Data Table 2; Supplementary Table S3).

**Theorem statement.**

**Definition 1** (Imaging Operator Class).  $\mathcal{C}_{\text{img}}$  consists of all imaging forward models expressible as a finite sequential-parallel composition of stages—each either linear with bounded operator norm ( $\|H_k\| \leq B$ ) or pointwise nonlinear with bounded Lipschitz constant ( $\text{Lip}(\Lambda_k) \leq L$ )—with stage count  $K \leq N_{\text{max}}$ .

**Definition 2** ( $\varepsilon$ -Approximate Representation). A typed DAG  $G$  with nodes  $V \subseteq \mathcal{B}$  is an  $\varepsilon$ -approximate representation of  $H \in \mathcal{C}_{\text{img}}$  if  $\sup_{\|\mathbf{x}\| \leq 1} \|H(\mathbf{x}) - H_G(\mathbf{x})\| / (\|H(\mathbf{x})\| + \delta) \leq \varepsilon$ , where  $\delta > 0$  is a regularization constant, and  $|V| \leq N_{\text{max}}$ ,  $\text{depth}(G) \leq D_{\text{max}}$ .

132 **Theorem 3** (Finite Primitive Basis). *For every  $H \in \mathcal{C}_{\text{img}}$ , there exists a typed DAG  $G$  with*  
 133  *$V \subseteq \mathcal{B}$  that is an  $\varepsilon$ -approximate representation of  $H$ .*

134 *Proof sketch.* By Definition 1,  $H = H_K \circ \dots \circ H_1$  with  $K \leq N_{\text{max}}$ . Each factor is classified  
 135 into one of six physics-stage families and realized by primitives: propagation  $\rightarrow P$  or  $C$ ; elas-  
 136 tic interaction  $\rightarrow M$ ; inelastic interaction  $\rightarrow R$ ; pointwise nonlinear physics  $\rightarrow \Lambda$ ; encoding-  
 137 projection  $\rightarrow \Pi$  or  $F$ ; detection-readout  $\rightarrow$  finite composition from  $\{\Sigma, S, W, C, D\}$ . For  
 138 linear stages, per-factor errors compose via sub-multiplicativity:  $\|H - H_G\| \leq K \cdot \max_k(\varepsilon_k) \cdot$   
 139  $B^{K-1}$ . For nonlinear stages, the Lipschitz composition bound  $\|H(\mathbf{x}) - \hat{H}(\mathbf{x})\| \leq \sum_k \varepsilon_k \prod_{j>k} \gamma_j$   
 140 applies, where  $\gamma_j$  is the Lipschitz constant of stage  $j$ . Both bounds yield  $\leq \varepsilon$  under the  
 141 stated regularity conditions. Full proof in Supplementary Note 12; formal primitive seman-  
 142 tics and typed DAG denotation in<sup>1</sup>. □

143 **Scope of  $\mathcal{C}_{\text{img}}$ .** Covers all clinical, scientific, and industrial imaging modalities—linear and  
 nonlinear—including coded aperture, interferometric, projection, Fourier-encoded, acoustic,  
 scattering, and strongly nonlinear or stochastic regimes (beam hardening, phase wrapping, mul-  
 tiple scattering, and stochastic transport operators often evaluated via Monte Carlo sampling).  
 Also includes quantum state tomography and relativistic measurement models. Full-wave and  
 stochastic-transport forward models are covered as Level-4 fidelity implementations, not as dis-  
 tinct modalities. No modality within the stated operator class is excluded; out-of-class cases are  
 handled by the extension protocol below. Formal definitions and proofs in<sup>1</sup>.

144 **Extension protocol.** A new primitive is warranted when no DAG over  $\mathcal{B}$  achieves  $e_{\text{img}} \leq$   
 145  $\varepsilon$  within the complexity bounds. The extension requires: (1) validated forward/adjoint,  
 146 (2) demonstrated representation gap, (3) error reduction below  $\varepsilon$ , (4) need by  $\geq 2$  modal-  
 147 ities, (5) backward-compatible closure re-test. Scatter ( $R$ ) was added via this protocol when  
 148 Compton imaging produced  $e_{\text{img}} = 0.34$  without it.

149 **Held-out closure test.** To validate Theorem 3 empirically, we conduct a closure test un-  
 150 der a frozen protocol: the primitive library  $\mathcal{B}$  (11 primitives), Detect families (5), Transform  
 151 families (5), fidelity threshold ( $\varepsilon = 0.01$ ), and complexity bounds ( $N_{\text{max}} = 20$ ,  $D_{\text{max}} = 10$ )  
 152 are all frozen before evaluation. We test 5 held-out modalities (OCT, photoacoustic, SIM,  
 153 phase-contrast X-ray, electron ptychography) and 3 exotic modalities (quantum ghost imag-  
 154 ing, THz-TDS, Compton scatter imaging):

155 Seven of eight modalities decompose with the frozen library. Quantum ghost imaging  
 156 is operator-equivalent to a single-pixel camera at the image-formation level—sharing the  
 157 canonical DAG Source  $\rightarrow M \rightarrow \Sigma \rightarrow D$  despite fundamentally different source physics.  
 158 THz-TDS requires only the coherent-field Detect family. Compton scatter imaging trig-  
 159 gered the extension protocol (Section “Extension protocol” above): its representation gap  
 160 ( $e_{\text{img}} = 0.34 \gg \varepsilon$ ) met the falsifiable criterion for basis incompleteness, and the disci-  
 161 plined resolution—adding Scatter ( $R$ )—covers 5+ additional modalities (Raman, fluores-  
 162 cence, DOT, Brillouin) while preserving all existing decompositions. The basis grows from

| Modality             | $e_{\text{img}}$ | #Nodes/Depth | Triad transfer | New prim.? |
|----------------------|------------------|--------------|----------------|------------|
| OCT                  | $< 0.01$         | 5 / 3        | Y              | N          |
| Photoacoustic        | $< 0.01$         | 4 / 3        | Y              | N          |
| SIM                  | $< 0.01$         | 4 / 3        | Y              | N          |
| Phase-contrast X-ray | $< 0.01$         | 5 / 4        | Y              | N          |
| Electron ptycho      | $< 0.01$         | 4 / 3        | Y              | N          |
| Ghost imaging        | $< 0.01$         | 4 / 3        | Y              | N          |
| THz-TDS              | $< 0.01$         | 3 / 2        | Y              | N          |
| Compton scatter      | $< 0.01^*$       | 4 / 3        | Y              | Y ( $R$ )  |

\*With  $R$  in library;  $e_{\text{img}} > 0.01$  without  $R$ .

Max observed: 5 nodes / depth 4; median: 4 nodes / depth 3 (cf. bounds  $N_{\text{max}} = 20$ ,  $D_{\text{max}} = 10$ ).

163 9 to 11 (22% increase) while modality coverage grows by 19%+.

164 **Basis-growth saturation.** Plotting the number of distinct primitives  $K$  against the num-  
165 ber of registered modalities  $N$  reveals clear saturation (Figure 7): 9 of 11 primitives are  
166 introduced by the first 10 modalities, Disperse ( $W$ ) by CASSI-type spectral systems, Scat-  
167 ter ( $R$ ) when Compton/Raman-class modalities enter, and Transform ( $\Lambda$ ) when nonlinear  
168 physics stages (beam hardening, phase wrapping) are encountered. The growth is sublinear  
169 and saturating:  $K = 11$  at  $N = 30$ , with no new primitive required for the most recent 18  
170 modalities added. This saturation is consistent with Theorem 3: once all six physics-stage  
171 families are covered by primitives, new modalities compose existing primitives rather than  
172 requiring new ones.

173 **Theorem tightness and minimality.** The theoretical complexity bounds ( $N_{\text{max}} = 20$ ,  
174  $D_{\text{max}} = 10$ ) are conservative: empirically, all 26 registered and 8 held-out modalities ( $\sim 30$   
175 unique, accounting for 4 overlaps) require  $\leq 6$  nodes and depth  $\leq 5$  (held-out closure test  
176 table above), with the median modality using 4 nodes at depth 3. The gap between empirical  
177 and theoretical bounds reflects the compactness of real physical imaging chains. Moreover,  
178 the basis is *minimal*: for each of the 11 primitives, we exhibit a witness modality whose  
179 representation error exceeds  $\varepsilon$  when that primitive is removed (Supplementary Note 10,  
180 Proposition 1 therein; beam hardening CT witnesses the necessity of Transform  $\Lambda$ ). Eight  
181 primitives ( $P$ ,  $M$ ,  $\Pi$ ,  $F$ ,  $\Sigma$ ,  $D$ ,  $R$ ,  $\Lambda$ ) are strictly necessary; the remaining three ( $C$ ,  $S$ ,  $W$ )  
182 are necessary under the stated complexity bounds.

## 183 The Triad Decomposition

184 The TRIAD DECOMPOSITION asserts that every failure in computational image recovery  
185 within  $\mathcal{C}_{\text{img}}$  can be attributed to one or more of exactly three root causes, which we term  
186 *gates*. The three gates are mutually exclusive in their physical origin yet may co-occur and  
187 interact in any given measurement scenario.

188 **Why exactly three gates.** The reconstruction pipeline has exactly three inputs: a  
 189 sensing geometry  $H$  that determines what information is captured (Gate 1), a physical  
 190 carrier whose statistics determine the noise floor (Gate 2), and a forward model  $H_{\text{nom}}$   
 191 that the solver inverts (Gate 3). Any reconstruction error must trace to one of these  
 192 inputs: information that was never captured (null space of  $H$ ), information that was cap-  
 193 tured but corrupted (carrier noise), or information that was captured but misinterpreted  
 194 (model error  $H_{\text{nom}} \neq H_{\text{true}}$ ). This trichotomy is exhaustive: the MSE decomposes as  
 195  $\text{MSE} \leq \text{MSE}^{(G1)} + \text{MSE}^{(G2)} + \text{MSE}^{(G3)}$  (Supplementary Note 1), with no residual term.  
 196 Solver suboptimality (e.g., insufficient iterations or regularization mismatch) is subsumed  
 197 by Gate 3 in the Triad formalism, because the solver’s implicit forward model includes its  
 198 regularization assumptions.

199 **Gate 1: Recoverability.** **Gate 1** asks whether the measurement encodes sufficient in-  
 200 formation about the signal of interest. Formally, if the forward operator  $H \in \mathbb{R}^{m \times n}$  maps  
 201 the unknown signal  $\mathbf{x} \in \mathbb{R}^n$  to the measurement  $\mathbf{y} = H\mathbf{x} + \mathbf{n}$ , then the null space  $\mathcal{N}(H)$   
 202 defines signal components that are fundamentally invisible to the sensor. When  $\mathcal{N}(H)$  is  
 203 large, no solver can recover the missing information. **Gate 1** failures are intrinsic to the  
 204 measurement design and can only be remedied by acquiring additional data.

205 **Gate 2: Carrier Budget.** **Gate 2** asks whether the signal-to-noise ratio (SNR) is suffi-  
 206 cient. Every physical carrier—photons, electrons, spins, acoustic waves, particles—is sub-  
 207 ject to fundamental noise limits: shot noise for photon-counting systems, thermal noise  
 208 in electronic detectors,  $T_1/T_2$  relaxation noise in magnetic resonance. When the carrier  
 209 budget is too low, the measurement is dominated by noise and the reconstruction degrades  
 210 regardless of operator fidelity.

211 **Gate 3: Operator Mismatch.** **Gate 3** asks whether the forward model assumed by the  
 212 solver matches the true physics. The solver operates with a nominal operator  $H_{\text{nom}}$ , but  
 213 the data were generated by a true operator  $H_{\text{true}}$ . When  $H_{\text{nom}} \neq H_{\text{true}}$ , the reconstruction  
 214 targets a phantom inverse problem. **Gate 3** failures are insidious because they produce  
 215 structured artifacts that mimic signal content, leading practitioners to blame the solver  
 216 rather than the model. Sources include geometric misalignment (mask shift, rotation),  
 217 parameter drift (coil sensitivity variation, gain instability), and model simplification.

218 **Definition 4** (Triad Decomposition). *Let  $H \in \mathcal{C}_{\text{img}}$  with measurement  $\mathbf{y} = H\mathbf{x} + \mathbf{n}$  and*  
 219 *solver using nominal operator  $H_{\text{nom}}$ . The reconstruction error decomposes as  $\text{MSE} \leq$*   
 220  *$\text{MSE}^{(G1)} + \text{MSE}^{(G2)} + \text{MSE}^{(G3)}$ , where  $\text{MSE}^{(G1)}$  is the null-space loss (Supplementary*  
 221 *Eq. S1),  $\text{MSE}^{(G2)}$  is the noise-floor term (Supplementary Eq. S3), and  $\text{MSE}^{(G3)}$  is the*  
 222 *mismatch-induced term (Supplementary Eq. S5).*



223 **Theorem 5** (Gate 3 Dominance). *For any  $H \in \mathcal{C}_{\text{img}}$  operating above its Gate 1 floor*  
 224 *( $\gamma > \gamma_{\text{min}}$ ) and Gate 2 floor ( $\text{SNR} > \text{SNR}_{\text{min}}$ ), Gate 3 is the binding constraint whenever*  
 225  *$\|\delta\theta\|_{\mathbf{J}^\top \mathbf{J}} > (\sigma_n / \|\mathbf{x}\|_\infty) \cdot \kappa(\mathbf{H})$ .*

226 *Proof.* See Supplementary Note 1, Proposition 2. □

227 **Key finding: Gate 3 dominates.** Across all 14 validated configurations (12 modalities  
 228 spanning 5 carrier families), **Gate 3** is the dominant failure gate ( $C_{\text{mismatch}} > \max(C_{\text{noise}}, C_{\text{recover}})$ )  
 229 in 14/14 cases; Supplementary Tables S12–S13 confirm that Gates 1–2 bind only at extreme  
 230 compression or photon starvation). In CASSI, a sub-pixel mask shift degrades MST-L by  
 231  $13.98 \pm 0.62$  dB (mean  $\pm$  s.d. over 10 scenes); in MRI, a 5% coil sensitivity mismatch under  
 232 clinically realistic multi-coil conditions (8 coils,  $4\times$  acceleration) degrades reconstruction  
 233 by 1.75–7.14 dB (Supplementary Note 11); in cryo-EM, a 200 nm defocus error degrades  
 234 Wiener-filter reconstruction by 1.1 dB; in ultrasound, a 200 m/s speed-of-sound error de-  
 235 grades DAS beamforming by 0.4 dB. Calibration—not solver design—is the underinvested  
 236 bottleneck: a single forward-model correction recovers more reconstruction quality than the  
 237 gap between traditional and state-of-the-art solvers.

238 **Relationship to the Finite Primitive Basis.** The two contributions are complemen-  
 239 tary: the Finite Primitive Basis provides a universal representation (every forward model  
 240 is a DAG over 11 primitives<sup>1</sup>), and the Triad provides a universal diagnostic law over that  
 241 representation. The DAG structure makes Gate 3 diagnosis *actionable*: the MismatchAgent  
 242 localizes the offending primitive node and corrects its parameters.

## 243 Consequences: Diagnosis and Correction

244 The two theoretical results directly imply a practical framework. The OPERATORGRAPH  
 245 provides a modality-agnostic representation; the Triad provides root-cause diagnosis. Three  
 246 deterministic agents—one per gate—evaluate information capacity, carrier budget, and op-  
 247 erator mismatch respectively, producing a quantitative TRIADREPORT for every imaging  
 248 configuration (see Methods for implementation details).

249 **Correction pipeline.** When **Gate 3** is dominant, a two-stage correction pipeline refines  
 250 the forward model parameters: a coarse beam search over the mismatch parameter family  
 251  $\theta = (\theta_1, \dots, \theta_k)$  associated with the offending DAG node, followed by gradient refinement  
 252 via backpropagation through the differentiable forward model. Critically, correction oper-  
 253 ates exclusively on the forward model parameters, not on the solver weights: any existing  
 254 solver—iterative, plug-and-play, or deep unrolling—benefits from the corrected operator  
 255 without modification or retraining.



**4-Scenario Protocol.** To rigorously evaluate correction quality, PWM defines four canonical scenarios. **Scenario I** (Ideal): the solver reconstructs using the true operator  $H_{\text{true}}$ . **Scenario II** (Mismatch): the solver uses the nominal operator  $H_{\text{nom}}$  on data generated by  $H_{\text{true}}$ . **Scenario III** (Corrected): the solver uses the PWM-corrected operator. **Scenario IV** (Oracle): the true operator on mismatched data, providing the correction ceiling. The recovery ratio  $\rho = (\text{PSNR}_{\text{III}} - \text{PSNR}_{\text{II}}) / (\text{PSNR}_{\text{I}} - \text{PSNR}_{\text{II}})$  quantifies how much of the mismatch-induced degradation is recovered (see Methods, Equation (5)).

## Empirical Validation

We validate both theoretical contributions in two stages: controlled simulation experiments across twelve modalities using the 4-Scenario Protocol, and hardware validation on real data from ten modalities spanning all five carrier families—incoherent photons (CASSI, CACTI), coherent photons (compressive holography, fluorescence microscopy), X-ray photons (CT, CBCT), electrons (ptychography, cryo-EM), nuclear spins (MRI), and acoustic waves (ultrasound). This is, to our knowledge, the broadest cross-carrier validation of any computational imaging framework.

### Simulation experiments

**Correction results.** Extended Data Table 1 (Supplementary Table S1) summarizes the oracle correction ceiling across 14 configurations spanning 12 modalities. The oracle gain  $\Delta_{\text{oracle}} = \text{PSNR}_{\text{IV}} - \text{PSNR}_{\text{II}}$  ranges from  $+0.76 \pm 0.12$  dB (CASSI/GAP-TV, 95% bootstrap CI over 10 scenes) to  $+10.68 \pm 0.41$  dB (CT) across the original photon/X-ray modalities. Autonomous grid-search calibration achieves 82–100% of the oracle ceiling (Supplementary Table S9). The validated modalities span photon-domain (CASSI:  $+0.76/+6.50$  dB [GAP-TV/MST-L]; CACTI:  $+10.21 \pm 0.85$  dB; SPC:  $+7.71/+10.38$  dB [FISTA-TV/HATNet]; Lensless:  $+3.55 \pm 0.22$  dB; compressive holography:  $+0.0/+0.4$  dB; fluorescence microscopy:  $+0.1/+6.8$  dB), coherent-photon (Ptychography:  $+7.09 \pm 0.47$  dB), X-ray (CT:  $+10.68 \pm 0.41$  dB; CT offset:  $+2.5/+3.9$  dB), acoustic (Ultrasound:  $+0.1/+0.9$  dB), and electron (Cryo-EM:  $+0.1/+3.6$  dB) and nuclear spin (MRI:  $+1.75$  to  $+7.14$  dB under clinically realistic multi-coil conditions at 3–5% sensitivity mismatch; Supplementary Note 11). All confidence intervals are computed via the bootstrap percentile method with  $B = 1,000$  resamples over test scenes (Methods). Correction gains are commensurate across all five carrier families, confirming carrier-agnostic operation.

**Modality deep dives.** In CASSI, a 5-parameter mismatch<sup>20</sup> collapses all solvers to  $\sim 21$  dB regardless of ideal performance (24–35 dB); oracle recovery varies by solver ( $\rho_{\text{IV}} = 0-0.57$ ; Supplementary Table S2), confirming that multi-parameter mismatches are harder to correct than isolated shifts. In CACTI, EfficientSCI<sup>21</sup> drops by 20.58 dB, yet GAP-TV

291 achieves 100% autonomous recovery of the oracle ceiling (Supplementary Table S9)—the  
 292 best ideal-condition solver suffers the largest degradation. SPC gain drift is corrected to  
 293 86–92% of the oracle ceiling (Table S9). **Gate 3** is dominant in every case; Gates 1–2 impose  
 294 irrecoverable information-theoretic limits only at extreme compression or photon starvation  
 295 (Supplementary Tables S12–S13).

296 **Zero-shot generalization.** Hyperparameters tuned on photon-domain modalities (CASSI,  
 297 CACTI, SPC) transfer without modification to coherent-photon, spin, and X-ray domains,  
 298 with comparable correction gains (Figure 4), confirming carrier-agnostic operation enabled  
 299 by the OperatorGraph abstraction.

## 300 Hardware validation

301 **CASSI real data.** These experiments use physical measurements from hardware-calibrated  
 302 coded aperture systems—not synthetic data—providing direct evidence that mismatch dom-  
 303 inance persists under real manufacturing tolerances and environmental conditions. We re-  
 304 construct 5 real TSA scenes<sup>12</sup> under calibrated and perturbed masks. GAP-TV shows a  
 305 mean residual ratio of  $1.8\times$  ( $0.00189 \rightarrow 0.00333$ ); HDNet shows  $1.0\times$  (mask-oblivious).  
 306 MST-S/L show ratios near  $1.0\times$  on real data, in contrast to severe synthetic degrada-  
 307 tion. This asymmetry is itself informative: real hardware already contains unmodeled  
 308 manufacturing imperfections, and an additional software-simulated perturbation is par-  
 309 tially absorbed by pre-existing calibration errors (Supplementary Table S7). An extended  
 310 cross-residual analysis (Supplementary Note 15) reveals a deeper diagnostic: the *cross-*  
 311 *residual*—evaluating a mismatched reconstruction under the true forward model—increases  
 312 monotonically from 0.3% at 0.25 px shift to 11.1% at 2.0 px shift, with per-scene standard  
 313 deviation  $< 0.4\%$ , confirming that Gate 3 sensitivity is scene-independent and monotonic  
 314 in mismatch magnitude. These real-data results independently confirm the Triad predic-  
 315 tions from simulation: the InverseNet benchmark<sup>20</sup> validates the same mismatch-dominance  
 316 pattern on physical DD-CASSI and CACTI instruments.

317 **CACTI real data.** GAP-TV shows  $10.4\times$  mean residual ratio under sub-pixel shift,  
 318 with per-scene ratios from  $9.4\times$  to  $11.0\times$ . An extended analysis (Supplementary Note 15)  
 319 reveals a striking dissociation: the *self-residual* barely changes ( $1.0$ – $1.12\times$ ) because the  
 320 solver adapts to the wrong model, but the *cross-residual* explodes  $44$ – $462\times$  as the shift  
 321 increases from 0.25 to 1.0 px. This demonstrates that self-consistency is not a reliable  
 322 diagnostic for operator mismatch in underdetermined systems, with direct implications for  
 323 compressive sensing quality control.

324 **CT real sinograms.** We extend hardware validation beyond optical instruments to three  
 325 additional carrier families. For **X-ray CT**, two public sinogram datasets—the FIPS walnut

micro-CT (1200 projections, 2296 detectors) and the Helsinki Tomography Challenge 2022 (721 projections, 560 detectors)—show that center-of-rotation (CoR) offsets of 1–8 pixels cause monotonic PSNR degradation of 9.4 dB (walnut) and 8.0 dB (HTC), with 100% oracle recovery (Supplementary Note 15). For **electron ptychography**, real 4D-STEM data from SrTiO<sub>3</sub> (Zenodo 5113449; 300 kV, 128 × 128 scan) shows that probe position jitter of 0.5–4.0 pixels causes monotonic iCoM phase degradation from 35.5 to 19.4 dB (16.1 dB total loss), with > 99.9% oracle recovery (Supplementary Note 15). For **MRI**, multi-coil brain  $k$ -space from M4Raw (Zenodo 8056074; 4 coils, 256 × 256) shows that coil sensitivity perturbation up to 40% degrades R=2 SENSE reconstruction by 0.5 dB, with 100% recovery via sensitivity re-estimation. The modest MRI effect is consistent with the over-determined encoding at R=2; the simulation experiments at R=4 (Supplementary Note 11) confirm that Gate 3 severity scales with acceleration factor.

**Ultrasound.** The acoustic carrier family—absent from all prior validation—is tested via simulated pulse-echo B-mode imaging with a 64-element linear array at 5 MHz centre frequency and 100 MHz sampling rate on a 128 × 128 tissue phantom. Speed-of-sound (SoS) mismatch of [50, 100, 200, 400] m/s (relative to true  $c = 1540$  m/s) causes monotonic PSNR degradation from 0.1 to 0.9 dB, with delay-and-sum (DAS) beamforming as the reconstruction solver. Grid-search calibration over 21 SoS candidates in [1440, 1640] m/s achieves  $\rho = 1.19$ –2.77 (Supplementary Table S14); the over-unity ratios indicate that the optimizer consistently finds an operating point ( $c = 1440$  m/s) that improves upon oracle reconstruction, reflecting the DAS beamformer’s inherent model simplifications. This confirms Gate 3 dominance for acoustic propagation, completing the five-carrier-family validation.

**Cryo-EM.** Electron-carrier validation uses simulated cryo-EM micrographs (128 × 128, pixel size 0.1 nm) with contrast transfer function (CTF) encoding. Defocus mismatch of [100, 200, 500, 1000] nm (relative to true  $\Delta f = -2000$  nm) degrades Wiener-filter reconstruction by 0.1–3.6 dB. At Nyquist frequency 5 nm<sup>-1</sup>, the CTF exhibits ~30 oscillations before the B-factor envelope ( $B = 2$  nm<sup>2</sup>) attenuates high frequencies, making the reconstruction highly sensitive to defocus error. Grid-search calibration over 51 defocus values in [−3000, −500] nm recovers  $\rho = 1.00$ –1.12 (Supplementary Table S15), confirming that CTF parameter mismatch is the dominant bottleneck in single-particle reconstruction—consistent with the extensive defocus-estimation literature in structural biology.

**CT detector offset.** X-ray CT validation uses a 128 × 128 Shepp–Logan phantom with 360-angle parallel-beam projections. Detector offset mismatch of [2, 5, 10, 20] pixels—simulated by shifting the sinogram along the detector axis—causes 2.5–3.9 dB PSNR degradation under filtered backprojection (FBP) with Shepp–Logan filter. At large offsets ( $\geq 10$  px), edge data loss additionally reduces the oracle PSNR from 13.8 to 10.6 dB (a Gate 1 effect).

Grid-search calibration over 21 offset candidates in  $[-25, 25]$  px achieves  $\rho = 0.83$ – $1.00$  (Supplementary Table S16), with exact offset recovery ( $\leq 0.5$  px error) for offsets  $\geq 5$  pixels. A  $512 \times 512$  scalability experiment with 720 projections confirms identical Gate 3 behaviour at clinical resolution: 1.0–4.1 dB degradation with  $\rho = 0.88$ – $1.00$ , demonstrating that mismatch dominance is resolution-invariant.

**Compressive holography.** Multi-depth Fresnel propagation encodes a 3D object ( $4 \times 64 \times 64$ ) into a single 2D hologram—the most compressive encoding among validated modalities ( $\lambda = 532$  nm, pixel pitch  $5 \mu\text{m}$ , depth spacing  $200 \mu\text{m}$ ). Propagation distance error of  $[10, 50, 100, 200] \mu\text{m}$  causes progressive defocus, degrading FISTA-TV reconstruction by up to 0.4 dB at  $200 \mu\text{m}$  error. Autonomous calibration via hologram residual minimisation ( $\|y - A_{\text{cal}}\hat{x}\|^2/\|y\|^2$ ) achieves  $\rho = 0.64$ – $1.59$  (Supplementary Table S17). The per-plane PSNR analysis reveals that deeper planes are more sensitive to distance error, consistent with the quadratic phase scaling of Fresnel kernels. The modest correction gains reflect the low sensitivity of inline holography to propagation distance error at this pixel pitch; off-axis holographic configurations with tighter fringe spacing would show stronger Gate 3 effects, consistent with Brady and Gehm’s analysis of compressive holographic sensing<sup>22</sup>.

**Fluorescence microscopy.** Dual-PSF Stokes-shift imaging—excitation and emission wavelengths produce distinct PSF widths—is validated under PSF sigma mismatch of  $[0.1, 0.3, 0.5, 1.0]$  pixels on a  $64 \times 64$  fluorescence phantom (1000 peak photons). Richardson–Lucy deconvolution (80 iterations) degrades by 0.1–6.8 dB. A 2D grid search over  $(\sigma_{\text{ex}}, \sigma_{\text{em}})$  with measurement-residual minimisation achieves  $\rho = 0.44$ – $0.75$  (Supplementary Table S18). Recovery is lowest at the smallest error ( $\Delta\sigma = 0.1$  px,  $\rho = 0.44$ ) where the calibration signal is near the noise floor, and strongest at moderate errors ( $\Delta\sigma = 0.5$  px,  $\rho = 0.75$ ) where the objective landscape is well-resolved.

**Autonomous calibration.** Grid-search calibration recovers  $\rho = 0.85$  (CASSI, 1,140 s; 95% CI  $[0.79, 0.91]$ ),  $\rho = 1.00$  (CACTI, 60 s; CI  $[0.97, 1.00]$ ),  $\rho = 0.86$ – $0.92$  (SPC via TV objective; CI width  $\leq 0.06$ ),  $\rho = 1.19$ – $2.77$  (Ultrasound SoS),  $\rho = 1.00$ – $1.12$  (Cryo-EM defocus),  $\rho = 0.83$ – $1.00$  (CT detector offset),  $\rho = 0.64$ – $1.59$  (compressive holography), and  $\rho = 0.44$ – $0.75$  (fluorescence PSF) of the oracle correction. Recovery ratios exceeding unity (ultrasound, cryo-EM, compressive holography) indicate that the optimizer found operating points marginally superior to the assumed ground-truth parameters, reflecting inherent model simplifications in the reconstruction algorithms. Single-parameter modalities (CACTI, CT) achieve near-perfect or above-unity recovery because their mismatch manifolds are low-dimensional with clean objective landscapes. CT CoR calibration on real sinograms also achieves  $\rho = 1.00$  (CI  $[0.99, 1.00]$ ). Multi-parameter modalities (CASSI, fluorescence) show lower but still substantial recovery, reflecting the higher-dimensional search space.

399 **Simulation-to-hardware gap.** CASSI shows smaller real degradation ( $1.8\times$ ) than pre-  
 400 dicted by simulation, because as-built masks already contain unmodeled imperfections that  
 401 absorb incremental perturbations. CACTI shows larger real sensitivity ( $10.4\times$ ) than simula-  
 402 tion, because the temporal mask pattern replicates errors across all compressed frames. This  
 403 bidirectional discrepancy—invisible without a unified cross-modality framework—carries a  
 404 methodological lesson: synthetic mismatch studies can both overestimate marginal impact  
 405 (CASSI) and underestimate cumulative sensitivity (CACTI).

## 406 Discussion

407 The central finding is that the space of imaging forward models has finite structural  
 408 complexity—11 primitives suffice and are necessary—and that operator mismatch, not in-  
 409 formation deficiency or noise, is the dominant reconstruction bottleneck across all validated  
 410 modalities and all five carrier families. The twelve validated modalities span the breadth  
 411 of modern imaging science: from coded aperture cameras (CASSI, CACTI) through clin-  
 412 ical scanners (CT, CBCT, MRI, ultrasound) to instruments at the frontier of structural  
 413 biology (cryo-EM, 2017 Nobel Prize in Chemistry) and super-resolution microscopy (flu-  
 414 orescence, 2014 Nobel Prize in Chemistry). In every case, the same 11 primitives and 3  
 415 gates apply. The implication is not that solver research is unnecessary, but that calibration  
 416 is systematically underinvested: a single forward-model correction step often recovers more  
 417 reconstruction quality than the gap between a classical solver and a state-of-the-art deep  
 418 network operating on the same mismatched operator.

419 **Implications for autonomous scientific discovery.** The bounded complexity of imag-  
 420 ing physics has consequences that extend beyond calibration. If every imaging forward  
 421 model—from a \$10 webcam to a particle accelerator beamline—decomposes into the same  
 422 11 primitives, then any autonomous system tasked with “learning to see” faces a *finite*  
 423 *search problem* over typed physical operators, not an open-ended function approximation.  
 424 This reframes the challenge for AI Scientists<sup>23</sup>: rather than discovering physical laws from  
 425 scratch, a machine learning system equipped with the OPERATORGRAPH representation  
 426 needs only to identify which subset of 11 known primitives is active in a new instrument  
 427 and estimate their parameters from data. The Triad Decomposition further constrains the  
 428 hypothesis space for failure: an AI Scientist confronting a degraded reconstruction has ex-  
 429 actly three root causes to consider, each with a formal test and a known correction strategy.  
 430 The observation that 9 of 11 primitives are introduced by the first 10 modalities, with the  
 431 final two (Disperse and Scatter) appearing only for exotic carrier physics, suggests that the  
 432 primitive manifold is dense at its core and sparse at its boundary—a structure confirmed by  
 433 the registry’s growth to 168 modalities without requiring a 12th primitive. This enables few-  
 434 shot generalization to new modalities from existing primitive combinations. More broadly,

the Finite Primitive Basis Theorem raises a question for computational science beyond imaging: if the forward model of every physical measurement system can be expressed in a small universal grammar, what analogous grammars exist for other physical processes, and what does their existence imply for the long-term trajectory of physics-informed machine learning?

**Implications for imaging system design.** The Finite Primitive Basis implies that imaging system design is a combinatorial optimization over 11 typed primitives, not an open-ended search. The OPERATORGRAPH representation makes this explicit: a designer specifies which primitives are active, their parameters, and their interconnection topology. The Triad Decomposition then provides a quantitative sensitivity analysis: for any candidate design, Gate 3 analysis predicts which parameters are most sensitive to mismatch, enabling tolerance-aware co-design of hardware and reconstruction algorithms. This is directly relevant to multi-scale camera architectures<sup>24</sup>, where dozens of sub-apertures must be jointly calibrated, and to snapshot compressive imaging systems<sup>8</sup>, where the measurement operator encodes the entire signal recovery problem.

**Implications for biology and medicine.** The framework has immediate relevance for the two largest imaging user communities. In structural biology, cryo-EM resolution is rate-limited by contrast transfer function (CTF) estimation—a textbook Gate 3 problem. A 200 nm defocus error at 300 kV limits the achievable resolution from  $\sim 2$  Å to  $\sim 3$  Å, and CTFFIND-class estimators require thousands of particles to converge. The PWM Triad diagnoses this as Gate 3 dominance with a 1-parameter correction manifold ( $\Delta f$ ), and autonomous defocus calibration recovers  $\rho = 1.00$ – $1.12$  of the oracle (Supplementary Table S15). In clinical imaging, CBCT geometric calibration drift is a recognised quality-assurance concern: AAPM Task Group 142<sup>25</sup> mandates monthly mechanical isocentre verification ( $\leq 2$  mm tolerance) and AAPM Task Group 66<sup>26</sup> specifies CT-simulator geometric accuracy requirements for treatment planning. Our validation shows that detector offsets of 5–20 pixels—commensurate with the mechanical tolerances flagged by these guidelines—cause 2.5–3.9 dB PSNR degradation with 100% oracle recovery, suggesting that automated Gate 3 monitoring could complement existing QA protocols. More broadly, ultrasound speed-of-sound inhomogeneity causes geometric distortion and resolution loss in abdominal imaging, and MRI coil sensitivity drift degrades parallel imaging reconstruction. All three failure modes are instances of Gate 3, and all three are correctable through the same forward-model calibration pipeline. The unification of these disparate calibration challenges under a single diagnostic framework suggests a path toward automated quality assurance across the clinical imaging fleet.

**Comparison with specialist calibration.** Specialist calibration methods—ESPIRiT<sup>10</sup> for MRI, ePIE<sup>11</sup> for ptychography, cross-correlation auto-focus for CT, CTFFIND for cryo-

EM—outperform PWM on their home modality when adequate calibration data is available (Supplementary Note S14). However, they degrade under data-limited conditions (ESPIRiT:  $-9.29$  dB at 24 ACS lines), require per-modality engineering, and do not exist for several validated modalities (CASSI, CACTI, compressive holography, ultrasound SoS correction). PWM’s value is *generality*: a single pipeline serving 12 modalities across all five carrier families—including two Nobel Prize-winning techniques and four clinical instruments—with robust performance across calibration data regimes. The two approaches are complementary—specialist estimates can initialize PWM correction, and PWM can refine specialist estimates.

**Hardware validation interpretation.** The hardware results reveal that Gate 3 sensitivity depends on the *condition number* of the encoding matrix. On real CASSI, perturbation produces smaller degradation ( $1.8\times$ ) than predicted, because the as-built mask already contains unmodeled manufacturing imperfections. On real CACTI, degradation is severe ( $10.4\times$ ), because the simpler temporal-mask optics lack this pre-existing error buffer. On real CT sinograms, CoR offset causes 8–9 dB loss; electron ptychography shows up to 16.1 dB phase degradation; ultrasound SoS mismatch causes 0.1–0.9 dB loss; cryo-EM defocus error causes up to 3.6 dB degradation; and CBCT detector offset produces 2.5–3.9 dB loss. This range—from sub-dB (well-conditioned MRI at  $R=2$ , ultrasound) to  $> 16$  dB (ptychography)—is itself a Triad prediction: the condition number of the encoding matrix determines mismatch sensitivity, with well-conditioned systems showing graceful degradation and ill-conditioned systems showing catastrophic failure. The extended cross-residual analysis further reveals that *self-consistency metrics are misleading* in underdetermined systems (CACTI cross-residual is  $462\times$  while self-residual is only  $1.12\times$ ), underscoring the need for forward-model-aware diagnostics.

**Boundary conditions and failure modes.** The framework has well-defined limitations that inform its scope of applicability:

- **Multi-parameter mismatch:** CASSI recovery ratios under 5-parameter mismatch are moderate ( $\rho = 22\text{--}46\%$ ), reflecting the inherent difficulty of simultaneously correcting multiple coupled parameters on a high-dimensional manifold. Single-parameter correction achieves 85–100%.
- **Nonlinear forward models:** The Triad diagnostic has been validated on linear forward models only; beam hardening CT and phase-wrapped MRI require additional validation (the 11-primitive basis formally covers these via Transform  $\Lambda^1$ ).
- **Undeclared mismatch:** Correction is limited to declared mismatch parameter families in the mismatch database; undeclared failure modes (e.g., nonlinear detector drift) require extending the database.



508 • **Software-perturbation protocol:** Current real-data experiments inject controlled  
 509 perturbations into calibrated measurements, isolating the effect of each mismatch  
 510 parameter. The CT, ptychography, and MRI experiments use genuinely independent  
 511 public datasets with unknown calibration states, providing complementary evidence.  
 512 Full hardware-in-the-loop validation with physical parameter displacement is specified  
 513 in Methods and Supplementary Note 8 and is underway.

514 **Falsifiable predictions.** The framework generates concrete, testable predictions:

- 515 1. **SIM:** Gate 3 should dominate when pattern phase error exceeds 0.05 rad. Predicted  
 516 correction: +3–8 dB.
- 517 2. **OCT:** Dispersion mismatch should produce +2–5 dB correction gain via the Disperse  
 518 primitive ( $W$ ).
- 519 3. **Electron ptychography:** Position errors exceeding 1/10 of the probe diameter  
 520 should trigger Gate 3 dominance. This prediction is *confirmed* by real 4D-STEM  
 521 validation: +5 to +16 dB correction on SrTiO<sub>3</sub> data (Supplementary Note 15).
- 522 4. **Cryo-EM:** CTF defocus mismatch of  $> 200$  nm should shift the FSC<sub>0.143</sub> resolution  
 523 threshold by  $> 0.5$  Å. Our simulation confirms 1.1–3.6 dB degradation at 200–1000 nm  
 524 defocus error.
- 525 5. **Ultrasound:** Speed-of-sound mismatch of  $> 100$  m/s should produce  $> 1$  mm geo-  
 526 metric distortion in B-mode images at clinical imaging depths. Our DAS simulation  
 527 confirms monotonic degradation of 0.1–0.9 dB.
- 528 6. **CBCT:** Geometric calibration drift of  $> 5$  pixels should produce cupping artifacts  
 529 with  $> 3$  dB degradation, consistent with the mechanical tolerance specifications of  
 530 AAPM TG-142<sup>25</sup> and TG-66<sup>26</sup>. Our FBP simulation confirms 3.9 dB loss at 5-pixel  
 531 detector offset with 100% oracle recovery.

532 These predictions are falsifiable: if any modality shows Gate 1 or Gate 2 dominance under  
 533 the stated conditions, the framework would require revision. Three of six predictions are now  
 534 confirmed by simulation or hardware data. Hardware-in-the-loop validation with physical  
 535 parameter displacement is the natural next step (Supplementary Note 8).

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**Author Contributions.** C.Y. conceived the project, designed the TRIAD DECOMPOSITION framework, proved the Finite Primitive Basis Theorem, developed the OPERATORGRAPH IR, implemented the agent and correction systems, performed all simulation and real-data experiments, and wrote the manuscript. X.Y. developed the GAP-TV reconstruction algorithm used as the primary solver across CASSI, CACTI, and SPC experiments, contributed the EfficientSCI architecture used for CACTI validation, provided the CASSI and CACTI forward model specifications and mismatch parameter characterizations that define the 5-parameter mismatch model, validated the real-data experimental protocols for both CASSI (TSA scenes) and CACTI instruments, and edited the manuscript.

**Competing Interests.** C.Y. is an employee of NextGen PlatformAI C Corp, which develops the PWM platform. The authors declare no other competing interests.

**Ethics Declarations.** This study does not involve human participants, human tissue, or animals. All experiments use publicly available benchmark datasets and simulated data.

**Data Availability.** All synthetic measurement data can be regenerated using the OPERATORGRAPH templates and mismatch parameters in the Supplementary Information. The KAIST hyperspectral dataset<sup>27</sup> and TSA real-data scenes used for CASSI experiments are publicly available. CACTI real-data scenes are available from the EfficientSCI repository<sup>21</sup>. CT sinograms are from the FIPS walnut dataset (Zenodo 1254206) and Helsinki Tomography Challenge 2022 (Zenodo 6984868). The 4D-STEM ptychography dataset (SrTiO<sub>3</sub> [001]) is from Zenodo 5113449. Multi-coil MRI  $k$ -space data (M4Raw) is from Zenodo 8056074. Ultrasound, cryo-EM, CBCT, compressive holography, and fluorescence microscopy experiments use procedurally generated phantoms with parameters and random seeds fully specified in Methods; all scripts are included in the repository.

**Code Availability.** The PWM codebase, including all OPERATORGRAPH templates, agent implementations, real-data validation scripts, and evaluation pipelines, is available at [https://github.com/integritynoble/Physics\\_World\\_Model](https://github.com/integritynoble/Physics_World_Model) under the PWM Noncommercial Share-Alike License v1.0 (see LICENSE in the repository).

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## Online Methods

### OperatorGraph Specification

**Formal definition.** The OPERATORGRAPH intermediate representation encodes the forward physics of any computational imaging modality as a directed acyclic graph (DAG)

572  $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ . Each node  $v_i \in \mathcal{V}$  wraps a *primitive operator* and implements two entry points:  
 573 `forward( $x$ )  $\rightarrow y$`  and `adjoint( $y$ )  $\rightarrow x$` , the latter defined only when the primitive is lin-  
 574 ear. Edges  $e_{ij} \in \mathcal{E}$  encode data flow: the output of node  $v_i$  is passed to node  $v_j$ . Each  
 575 node additionally exposes a set of learnable parameters  $\theta_i$  that may be perturbed during  
 576 mismatch simulation or optimized during calibration, as well as read-only metadata flags  
 577 (`is_linear`, `is_stochastic`, `is_differentiable`). The graph is stored as a declarative  
 578 YAML specification (`OperatorGraphSpec`) and compiled to an executable `GraphOperator`  
 579 object by the `GraphCompiler`.

580 **Node types.** Primitive operators fall into two categories:

- 581 • **Linear operators.** Convolution (`conv2d`), mask modulation (`mask_modulate`), sub-  
 582 pixel shift (`subpixel_shift_2d`), Radon transform (`radon_fanbeam`), Fourier encod-  
 583 ing (`fourier_encode`), spectral dispersion (`spectral_disperse`), Fresnel propagation  
 584 (`fresnel_propagate`), random projection (`random_project`), and structured illumi-  
 585 nation (`sim_modulate`). Each implements both `forward()` and `adjoint()`.
- 586 • **Nonlinear operators.** Squared magnitude (`magnitude_sq`), Poisson–Gaussian noise  
 587 (`poisson_gaussian`), saturation clipping (`saturation_clip`), phase retrieval nonlin-  
 588 earity (`phase_abs`), and detector quantization (`quantize`). These set `is_linear` =  
 589 `False` and raise `NotImplementedError` on `adjoint()`, except where a well-defined  
 590 pseudo-adjoint exists (*e.g.*, the identity adjoint for magnitude-squared in Gerchberg–  
 591 Saxton-type algorithms).

592 **Adjoint consistency and Lipschitz bounds as mathematical safety rails.** Two  
 593 structural constraints distinguish the OPERATORGRAPH IR from standard deep-learning  
 594 forward models and prevent either nonlinear primitive family from becoming a universal  
 595 approximator.

596 First, *adjoint consistency*: correctness of every linear primitive is verified by a ran-  
 597 domized dot-product test. For a primitive  $A$  with forward map  $A : \mathbb{R}^n \rightarrow \mathbb{R}^m$ , we draw  
 598  $x \sim \mathcal{N}(0, I_n)$  and  $y \sim \mathcal{N}(0, I_m)$  and compute

$$\delta = \frac{|\langle A^*y, x \rangle - \langle y, Ax \rangle|}{\max(|\langle A^*y, x \rangle|, \epsilon)} \quad (1)$$

599 where  $\epsilon = 10^{-12}$  guards against division by zero. The test is repeated  $n_{\text{trials}} = 5$  times  
 600 with independent random draws; the primitive passes if  $\delta_{\text{max}} < 10^{-6}$ . At the graph level,  
 601 a compiled `GraphOperator` composed entirely of linear nodes executes the same test over  
 602 the composed forward–adjoint chain. A `GraphAdjointCheckReport` records  $n_{\text{trials}}$ ,  $\delta_{\text{max}}$ ,  
 603 and  $\bar{\delta}$  for audit. All graph templates that consist solely of linear primitives pass this check.  
 604 Adjoint consistency is not merely a numerical correctness guarantee: it is a *physical faithful-*  
 605 *ness certificate*. A model whose adjoint is inconsistent implements a physically impossible

energy accounting, making gradient-based calibration unreliable and certifiable reconstruction bounds vacuous. Neural network forward models lack this certificate by construction, since their weight matrices have no physical meaning and their transposes are not computed or validated.

Second, *Lipschitz bounds*: the two nonlinear primitive families, Detect  $D$  and Transform  $\Lambda$ , are each restricted to five canonical function families with at most two scalar parameters. This restriction enforces bounded Lipschitz constants ( $\text{Lip}(\Lambda_k) \leq L$  for the explicit  $L$  derived in<sup>1</sup>) and prevents either primitive from becoming a universal approximator in the sense of Cybenko<sup>28</sup>. Concretely: a generic neural network layer with sigmoid activations can approximate any continuous function on a compact domain; neither Detect nor Transform can, because their function families are closed under composition only within the five enumerated types. This non-universality is the key property that makes the 11-primitive library *falsifiable*: if a modality’s forward physics cannot be represented within the restricted families, the extension protocol (Main Text, “Extension protocol”) fires and a new primitive must be justified physically. The combination of adjoint consistency and bounded Lipschitz nonlinearity provides the mathematical safety rails that make OPERATORGRAPH-based reconstruction both interpretable and certifiable—properties that standard physics-informed neural networks<sup>29</sup> achieve only approximately and without modality-agnostic guarantees.

**Graph compilation.** The compiler executes a four-stage pipeline:

1. **Validate.** Confirm acyclicity via topological sort (Kahn’s algorithm), verify that every `primitive_id` exists in the global `PRIMITIVE_REGISTRY`, reject duplicate `node_id` values, and optionally verify shape compatibility along edges when a `canonical_chain` metadata flag is set.
2. **Bind.** Instantiate each primitive with its parameter dictionary  $\theta_i$ .
3. **Plan forward.** The topological sort yields a sequential execution plan  $(v_{\pi(1)}, \dots, v_{\pi(|V|)})$ .
4. **Plan adjoint.** For graphs where `all_linear = True`, the adjoint plan reverses the topological order and applies each node’s individual adjoint in sequence, implementing the chain rule  $A^* = A_1^* \circ \dots \circ A_{|V|}^*$  for a composition  $A = A_{|V|} \circ \dots \circ A_1$ . For graphs containing nonlinear nodes, the adjoint plan is not generated, and any call to `adjoint()` raises `NotImplementedError` at runtime.

The compiled `GraphOperator` is serializable to JSON and hashable via SHA-256 for provenance tracking in RunBundle manifests.

**Template library.** The `graph_templates.yaml` registry contains 90 templates spanning all five carrier families. Of these, 26 modalities carry formal validation status (7 with full end-to-end correction validation, 1 with Scenario I baseline, 18 with template-level validation; Supplementary Table S3). Representative modalities by carrier family include:

- 642 • **Photons (optical and X-ray):** CASSI, SPC, CACTI, structured illumination  
643 microscopy (SIM), confocal, light-sheet, holography, ptychography, Fourier ptycho-  
644 graphic microscopy (FPM), optical coherence tomography (OCT), lensless imaging,  
645 light field, integral imaging, neural radiance fields (NeRF), Gaussian splatting, fluo-  
646 rescence lifetime imaging (FLIM), diffuse optical tomography (DOT), phase retrieval,  
647 X-ray computed tomography (CT), and cone-beam CT (CBCT).
- 648 • **Electrons:** Electron diffraction, electron backscatter diffraction (EBSD), electron  
649 energy loss spectroscopy (EELS), and electron holography.
- 650 • **Spins (MRI):** Functional MRI (fMRI), diffusion-weighted MRI (DW-MRI), and  
651 magnetic resonance spectroscopy (MRS).
- 652 • **Acoustic:** Ultrasound B-mode, Doppler ultrasound, shear-wave elastography, sonar,  
653 and photoacoustic tomography (combines optical excitation with acoustic detection).
- 654 • **Particles:** Neutron tomography, proton radiography, and muon tomography.

655 **Fidelity levels.** Each template is parameterized by a fidelity level that controls the degree  
656 of physical realism in the simulated forward model:

657 **Level 1 (Linear, shift-invariant):** The forward model is a linear, spatially uniform operator—  
658 the simplest approximation, suitable for initial diagnostics and rapid prototyping.

659 **Level 2 (Linear, shift-variant):** Spatially varying operator parameters (e.g. non-uniform  
660 illumination, position-dependent PSF, multi-coil sensitivity maps in MRI). Adds a  
661 modality-appropriate noise model (Poisson shot noise plus Gaussian read noise for  
662 photon-counting modalities, Rician noise for MRI, Poisson for CT).

663 **Level 3 (Nonlinear, ray/wave-based):** Includes nonlinear effects such as beam hard-  
664 ening, phase wrapping, and scattering, captured by the Transform  $\Lambda$  primitive. Per-  
665 turbation families and ranges are specified in `mismatch_db.yaml`.

666 **Level 4 (Full-wave / Monte Carlo):** Complete physical simulation including wave-optical  
667 propagation, spatially varying aberrations, detector nonlinearities, and environmental  
668 drift. Currently implemented for holography and ptychography.

669 All 11 primitives in the library  $\mathcal{B}$  apply uniformly across every fidelity level; the levels  
670 organize simulation complexity, not theorem scope.

## 671 Triad Decomposition Formalization

672 The TRIAD DECOMPOSITION asserts that the quality of any computational imaging re-  
673 construction is bounded by three fundamental gates. Rather than a qualitative guideline,

PWM quantifies each gate numerically and uses the resulting scores to diagnose the dominant bottleneck in any imaging configuration.

**Gate 1 (Recoverability).** Recoverability measures the information-theoretic capacity of the sensing geometry. We quantify it via the *effective compression ratio*  $r = m/n$ , where  $m$  is the number of independent measurements and  $n$  the dimension of the signal. The `compression_db.yaml` registry (1,186 lines) stores, for each modality, a lookup table mapping compression ratio to expected reconstruction PSNR under ideal conditions, obtained from calibration experiments or published benchmarks. Each entry carries a **provenance** field citing the source (paper DOI, internal experiment ID, or theoretical formula). Additional recoverability indicators include the effective rank of the measurement matrix (estimated via randomized SVD for large operators), the dimension of the null space, and the restricted isometry property (RIP) constant where analytically tractable (*e.g.*, for Gaussian random projections in SPC).

**Gate 2 (Carrier Budget).** The carrier budget quantifies the signal-to-noise ratio (SNR) of the measurement channel. The **PhotonAgent** consumes the `photon_db.yaml` registry (624 lines) which stores, per modality, a deterministic photon model parameterized by source power, quantum efficiency, exposure time, and detector characteristics. The agent classifies the noise regime into one of three categories: *shot-limited* (Poisson-dominated,  $\text{SNR} \propto \sqrt{N_{\text{photon}}}$ ), *read-limited* (Gaussian read noise dominates,  $\text{SNR} \propto N_{\text{photon}}/\sigma_{\text{read}}$ ), and *dark-current-limited* (long exposures where dark current accumulation dominates). The output is a **PhotonReport** containing the estimated SNR in decibels, the noise regime classification, per-element photon count, and a feasibility verdict (**sufficient**, **marginal**, or **insufficient**).

**Gate 3 (Operator Mismatch).** Operator mismatch quantifies the discrepancy between the assumed forward model  $H_{\text{nom}}$  and the true physical operator  $H_{\text{true}}$ . The **MismatchAgent** consults `mismatch_db.yaml` (797 lines) which catalogs, for each modality, the set of mismatch parameters (spatial shifts, rotational offsets, dispersion errors, PSF deviations, coil sensitivity errors, center-of-rotation offsets, *etc.*), their typical ranges, and available correction methods. The mismatch severity score  $s \in [0, 1]$  is computed as the normalized  $\ell_2$  distance  $\|\theta_{\text{true}} - \theta_{\text{nom}}\|/\|\theta_{\text{range}}\|$ , where  $\theta_{\text{range}}$  is the per-parameter dynamic range from the registry. Sensitivity analysis  $\partial\text{PSNR}/\partial\theta_k$  is estimated via finite differences on the forward model. The output is a **MismatchReport** containing the severity score, the dominant mismatch parameter, the recommended correction method, and the expected PSNR gain from correction.

**Gate binding determination.** Given reconstruction results under the four-scenario protocol (the Evaluation Protocol section below), PWM identifies the dominant gate by com-

710 paring three cost terms:

$$C_{\text{mismatch}} = \text{PSNR}_{\text{I}} - \text{PSNR}_{\text{II}} \quad (2)$$

$$C_{\text{noise}} = \text{PSNR}_{\text{ideal}} - \text{PSNR}_{\text{noisy}} \quad (3)$$

$$C_{\text{recover}} = \text{PSNR}_{\text{limit}} - \text{PSNR}_{\text{I}} \quad (4)$$

711 where  $\text{PSNR}_{\text{I}}$  is the reconstruction PSNR under Scenario I (ideal operator),  $\text{PSNR}_{\text{II}}$  under  
 712 Scenario II (mismatched operator),  $\text{PSNR}_{\text{noisy}}$  under the corresponding noisy condition,  
 713 and  $\text{PSNR}_{\text{limit}}$  is the theoretical upper bound from the compression table. The dominant  
 714 gate is  $\arg \max_g C_g$ .

715 **TriadReport.** For every diagnosis, PWM produces a TRIADREPORT: a Pydantic-validated  
 716 structured artifact comprising `dominant_gate` (enum: `recoverability`, `carrier_budget`,  
 717 `operator_mismatch`), `evidence_scores` (three floats, one per gate), `confidence_interval`  
 718 (float, 95% CI width from bootstrap), `recommended_action` (string, *e.g.* “increase compres-  
 719 sion ratio” or “apply mismatch correction”), and `parameter_sensitivities` (dictionary  
 720 mapping each mismatch parameter name to its  $\partial \text{PSNR} / \partial \theta_k$  value). The TRIADREPORT is  
 721 mandatory—PWM does not permit a reconstruction to be reported without an accompany-  
 722 ing diagnosis.

723 **Recovery ratio.** We define the *recovery ratio*

$$\rho = \frac{\text{PSNR}_{\text{III}} - \text{PSNR}_{\text{II}}}{\text{PSNR}_{\text{I}} - \text{PSNR}_{\text{II}}} \quad (5)$$

724 which lies in  $[0, 1]$  under standard convexity conditions (see Supplementary Note 1 for  
 725 formal analysis; values  $\rho > 1$  are possible when the corrected operator provides beneficial  
 726 regularization).  $\rho = 0$  indicates that calibration yields no benefit (mismatch is not the  
 727 bottleneck), while  $\rho = 1$  indicates that calibration fully closes the mismatch gap.

## 728 Agent System Architecture

729 The PWM agent system comprises 6 specialist agents, 1 optional hybrid agent, and 8  
 730 support classes totalling 10,545 lines of Python. All agents execute deterministically; no  
 731 large language model (LLM) is required for pipeline operation.

732 **PlanAgent.** The orchestrator agent. Given a user prompt or a structured `ExperimentSpec`,  
 733 PlanAgent parses the intent (`simulate`, `operator_correction`, or `auto`), maps the re-  
 734 quested modality to its canonical key via the `modalities.yaml` registry (which contains 168  
 735 modality entries across 19 categories with keywords, forward model equations, and default  
 736 solvers), builds an `ImagingSystem` contract, and dispatches to the appropriate sub-agents.



737 When the mode is `auto`, PlanAgent inspects the available data and operator specification  
738 to determine whether simulation or operator correction is more appropriate.

739 **PhotonAgent.** Computes SNR feasibility deterministically from the `photon.db.yaml`  
740 registry. For each modality and photon-level tier (`bright`, `standard`, `low_light`), the agent  
741 evaluates the photon budget by combining source power, quantum efficiency, exposure time,  
742 and noise model parameters. The output `PhotonReport` is a strict Pydantic model contain-  
743 ing `noise_regime` (enum), `snr_db` (float), `feasibility` (enum), and `per_element_photons`  
744 (float).

745 **RecoverabilityAgent.** A table-driven agent that consults `compression.db.yaml` (1,186  
746 lines) to map the modality and compression ratio to an expected PSNR range. Each table  
747 entry includes provenance metadata citing the original source. The output `RecoverabilityReport`  
748 contains `compression_ratio`, `psnr_prediction`, `feasibility`, and `null_space_dim` where  
749 available.

750 **MismatchAgent.** Scores the mismatch severity for a given imaging configuration us-  
751 ing `mismatch.db.yaml` (797 lines). For each modality, the database enumerates the rel-  
752 evant mismatch parameters, their physical units, typical perturbation ranges, and avail-  
753 able correction algorithms. The output `MismatchReport` includes `severity` (float, 0–1),  
754 `correction_method` (string), `expected_gain_db` (float), and `dominant_parameter` (string).

755 **AnalysisAgent.** The bottleneck classifier. It receives reports from the Photon, Recover-  
756 ability, and Mismatch agents, computes the gate costs (Equations (2) to (4)), identifies the  
757 dominant gate, and generates actionable suggestions. The AnalysisAgent also computes  
758 the recovery ratio  $\rho$  and its bootstrap confidence interval.

759 **AgentNegotiator.** Implements a cross-agent veto protocol. Before reconstruction is au-  
760 thorized, the negotiator inspects all three upstream reports and applies three veto con-  
761 ditions: (1) low photon budget combined with aggressive compression ( $C_{\text{noise}}$  and  $C_{\text{recover}}$   
762 both large); (2) severe mismatch ( $\text{severity} > 0.7$ ) without a planned correction step; (3) joint  
763 probability below the floor threshold ( $p_{\text{joint}} < 0.15$ ), indicating that all three subsystems  
764 are simultaneously marginal. When any veto fires, reconstruction halts with an actionable  
765 explanation and suggested remediation.

766 **HybridAgent.** An optional wrapper that invokes an LLM for natural-language narra-  
767 tive generation or edge-case modality mapping. All quantitative decisions remain on the  
768 deterministic code path; the HybridAgent is never required for pipeline operation.

**Support classes.** The remaining components include: `AssetManager` (file I/O and caching for large arrays), `ContinuityChecker` (verifies that sequential pipeline outputs are dimensionally consistent), `SystemDiscern` (auto-detects modality from uploaded data), `PreflightChecker` (validates the complete experiment configuration before execution), `WhatIfPrecomputer` (evaluates counterfactual what-if scenarios), `SelfImprovement` (logs diagnostic events for future registry refinement), `PhysicsStageVisualizer` (generates intermediate visualizations at each pipeline stage), and `UPWMI` (Universal Physics World Model Interface, the top-level entry point that wires all agents together).

**Contract system.** Inter-agent communication uses 25 Pydantic v2 contract models. All contracts inherit from `StrictBaseModel`, which enforces `extra="forbid"` (no unexpected fields), `validate_assignment=True` (mutations re-validated), and a model validator that rejects NaN and Inf in any float field. Bounded scores use `Field(ge=0.0, le=1.0)`. Enums are string enums for human-readable JSON serialization. This design ensures that pipeline failures surface immediately as validation errors rather than propagating silently.

**YAML registries.** The system is driven by 9 YAML registries totalling 7,034 lines: `modalities.yaml` (modality definitions), `graph_templates.yaml` (OperatorGraph skeletons), `photon_db.yaml` (photon models), `mismatch_db.yaml` (mismatch parameters and correction methods), `compression_db.yaml` (recoverability tables with provenance), `solver_registry.yaml` (solver configurations), `primitives.yaml` (primitive operator metadata), `dataset_registry.yaml` (dataset locations and formats), and `acceptance_thresholds.yaml` (pass/fail thresholds per metric).

## Correction Algorithms

We implement two complementary algorithms for operator mismatch correction. Crucially, both algorithms operate on the forward operator parameters  $\theta$  rather than the reconstruction solver weights, making them *solver-agnostic*: the corrected operator  $H(\hat{\theta})$  benefits any downstream solver (GAP-TV, MST-L, HDNet<sup>30</sup>, CST, *etc.*) without retraining.

**Algorithm 1: Hierarchical Beam Search.** The coarse correction phase employs a hierarchical search strategy to rapidly explore the mismatch parameter space. For CASSI, the five-parameter mismatch model comprises mask affine parameters (spatial shifts  $dx$ ,  $dy$  and rotation  $\theta$ ) and dispersion parameters (slope  $a_1$  and axis angle  $\alpha$ ); an optional sixth parameter, PSF width  $\sigma_{\text{psf}}$ , is available but not used in the primary experiments. The algorithm proceeds as follows:

1. **1D sweeps.** Each parameter is swept independently over its full range while holding others at nominal values. This produces five 1D cost curves from which coarse optima are extracted.

- 804    2. **3D beam search.** The mask affine subspace  $(dx, dy, \theta)$  is searched over a  $5 \times 5 \times 5$   
805       grid centered on the 1D optima. The top- $k$  ( $k = 5$ ) candidates by reconstruction  
806       PSNR are retained.
- 807    3. **2D beam search.** For each retained mask candidate, the dispersion subspace  $(a_1, \alpha)$   
808       is searched over a  $5 \times 7$  grid. The joint top- $k$  candidates are retained.
- 809    4. **Coordinate descent refinement.** Three rounds of univariate refinement on each  
810       parameter, shrinking the search interval by factor 2 at each round, produce the final  
811       estimate  $\hat{\theta}_{\text{Alg1}}$ .

812       Total runtime is approximately 300 seconds per scene on a single GPU. Accuracy is  
813        $\pm 0.1$ – $0.2$  pixels for spatial parameters and  $\pm 0.05^\circ$  for angular parameters.

814    **Algorithm 2: Joint Gradient Refinement.** The fine correction phase uses a differen-  
815       tiable forward model to jointly optimize all mismatch parameters via gradient descent. The  
816       key components are:

- 817    1. **Differentiable mask warp.** The binary mask is warped by a continuous affine  
818       transformation using bilinear interpolation, implemented as a custom PyTorch module  
819       (`DifferentiableMaskWarpFixed`). The mask values are passed through a straight-  
820       through estimator (STE) to maintain binary structure while permitting gradient flow.
- 821    2. **Differentiable forward model.** The CASSI forward model  $y = \text{CASSI}(x; \theta)$  is  
822       implemented as a differentiable PyTorch module (`DifferentiableCassiForwardSTE`)  
823       that accepts mismatch parameters as differentiable inputs.
- 824    3. **GPU grid initialization.** A full-range 3D grid search over  $(dx, dy, \theta)$  with  $9 \times 9 \times 7 =$   
825       567 points provides diverse starting candidates. The top 9 candidates seed multi-start  
826       gradient refinement.
- 827    4. **Staged gradient refinement.** Each of the 9 candidates is refined using Adam  
828       optimization (learning rate  $10^{-2}$ , decaying to  $10^{-3}$ ) for 200 steps. For each candidate,  
829       4 random restarts with jittered initialization guard against local minima. The loss  
830       function is the negative PSNR computed via an unrolled  $K$ -iteration differentiable  
831       GAP-TV solver (`DifferentiableGAPTV`,  $K = 10$  unrolled iterations).

832       Total runtime for Algorithm 2 is approximately 3,200 seconds (200 steps  $\times$  4 restarts  $\times$   
833       9 candidates with early stopping). Accuracy improves to  $\pm 0.05$ – $0.1$  pixels, a 3–5 $\times$  improve-  
834       ment over Algorithm 1. The two algorithms are used sequentially in practice: Algorithm 1  
835       provides a warm start, and Algorithm 2 refines to sub-pixel precision.

## Evaluation Protocol

**Four-Scenario Protocol.** We evaluate every modality under four standardized scenarios that isolate different sources of quality degradation:

**Scenario I (Ideal):**  $\mathbf{y}_{\text{obs}} = H_{\text{true}} \mathbf{x}_{\text{gt}}$ ; reconstruct with  $H_{\text{true}}$ . In this scenario the system is perfectly calibrated ( $H_{\text{true}} = H_{\text{nom}}$ ), so the operator used for reconstruction matches the one that generated the data. This yields the oracle upper bound on reconstruction quality, limited only by the sensing geometry and solver convergence.

**Scenario II (Mismatch):**  $\mathbf{y}_{\text{obs}} = H_{\text{true}} \mathbf{x}_{\text{gt}}$ ; reconstruct with  $H_{\text{nom}}$  ( $H_{\text{nom}} \neq H_{\text{true}}$ ). This is the standard operating condition in practice: the measurement is generated by the true physics, but the reconstruction uses a nominal (potentially mismatched) forward model.

**Scenario III (Corrected):**  $\mathbf{y}_{\text{obs}} = H_{\text{true}} \mathbf{x}_{\text{gt}}$ ; reconstruct with  $\hat{H} = H(\hat{\theta})$  where  $\hat{\theta}$  is estimated by Algorithms 1 and 2. This quantifies the benefit of mismatch calibration.

**Scenario IV (Oracle Mask):** Same measurements as Scenario II ( $\mathbf{y}_{\text{obs}} = H_{\text{true}} \mathbf{x}_{\text{gt}}$  with  $H_{\text{true}} \neq H_{\text{nom}}$ ); reconstruct with  $H_{\text{true}}$  instead of  $H_{\text{nom}}$ . Provides the correction ceiling: the best reconstruction achievable when the true operator is known exactly, applied to data that were sensed by the mismatched system. The gap between Scenario IV and Scenario I reveals the irreducible loss from the degraded sensing configuration itself (e.g., a shifted mask pattern is suboptimal even when perfectly known).

**Metrics.** Reconstruction quality is assessed using three complementary metrics:

- **PSNR** (peak signal-to-noise ratio, in dB): the primary metric, computed per scene and averaged. For signals normalized to  $[0, 1]$ ,  $\text{PSNR} = 10 \log_{10}(1/\text{MSE})$ . For SPC data normalized to  $[0, 255]$ , the peak value is 255.
- **SSIM** (structural similarity index): captures perceptual quality including luminance, contrast, and structural components, computed with a Gaussian window of width 11 and standard deviation 1.5.
- **SAM** (spectral angle mapper): for hyperspectral modalities (CASSI), measures the angle between predicted and true spectral vectors at each spatial location, reported in degrees. Lower is better.

## Datasets.

- **CASSI:** 10 scenes from the KAIST dataset<sup>27</sup>, each a  $256 \times 256 \times 28$  spectral cube (28 spectral bands from 450 nm to 650 nm). Data range  $[0, 1]$ .

868 • **CACTI**: 6 benchmark videos, each  $256 \times 256 \times 8$  (8 temporal frames encoded per  
869 snapshot). Data range  $[0, 1]$ .

870 • **SPC**: 11 natural images from the Set11 benchmark, each  $256 \times 256$  grayscale. Data  
871 range  $[0, 255]$ .

872 All per-scene metrics are reported individually as well as averaged, and all reconstruction  
873 arrays are saved as NumPy NPZ files.

## 874 Experimental Details

875 **Hardware.** All experiments are conducted on a single NVIDIA GPU. Algorithm 1 (beam  
876 search) and all solver-based reconstructions use the GPU for matrix–vector products and  
877 FFT operations. Algorithm 2 (gradient refinement) additionally uses PyTorch automatic  
878 differentiation on the same GPU.

879 **CASSI configuration.** The coded aperture snapshot spectral imaging (CASSI) system  
880 uses a TSA-Net binary mask of dimensions  $256 \times 256$ , with 28 spectral bands dispersed along  
881 the spatial dimension. The five-parameter mismatch model  $\theta = (dx, dy, \theta, a_1, \alpha)$  describes:  
882 mask spatial shift in  $x$  ( $dx$ , pixels), mask spatial shift in  $y$  ( $dy$ , pixels), mask rotation angle  
883 ( $\theta$ , degrees), dispersion slope ( $a_1$ , pixels per band), and dispersion axis angle ( $\alpha$ , degrees).  
884 An optional sixth parameter, PSF blur width ( $\sigma_{\text{psf}}$ , pixels), is available but not used in the  
885 primary experiments. These mismatch parameter values were determined through system-  
886 atic characterization of realistic CASSI assembly errors (Supplementary Note 8). The true  
887 mismatch parameters are  $\theta_{\text{true}} = (dx = 0.5 \text{ px}, dy = 0.3 \text{ px}, \theta = 0.1^\circ, a_1 = 2.02, \alpha =$   
888  $0.15^\circ)$ . Solvers evaluated include TwIST<sup>31</sup>, GAP-TV<sup>32</sup>, DGSMP<sup>33</sup>, MST-L<sup>7</sup>, and CST-  
889 L<sup>34</sup>, all of which receive the same operator and differ only in their reconstruction algorithm.  
890 The supplementary per-scene analysis additionally includes DeSCI<sup>35</sup> and HDNet<sup>30</sup>.

891 **CACTI configuration.** The coded aperture compressive temporal imaging system uses  
892 binary temporal masks of dimensions  $256 \times 256$ , encoding 8 video frames into a single  
893 snapshot measurement. Mismatch is parameterized as a temporal mask timing offset (sub-  
894 frame shift). The default solver is EfficientSCI<sup>21</sup>.

895 **SPC configuration.** The single-pixel camera uses random binary measurement patterns  
896 at three compression ratios: 10%, 25%, and 50% ( $r = m/n \in \{0.10, 0.25, 0.50\}$ ). Mismatch  
897 is modeled as an exponential gain drift ( $g_i = \exp(-\alpha \cdot i)$ ) on the measurement matrix. The  
898 default solver is FISTA-TV with total-variation regularization.

899 **MRI configuration.** Cartesian  $k$ -space sampling with  $4\times$  acceleration (25% of  $k$ -space  
900 lines acquired). Mismatch is parameterized as a 5% multiplicative error in the coil sensitivity

901 maps used for parallel imaging reconstruction. The default solver is SENSE<sup>17</sup> with  $\ell_1$ -  
 902 wavelet regularization.

903 **ESPIRiT comparison protocol.** To quantitatively compare PWM with ESPIRiT<sup>10</sup>,  
 904 we evaluate four conditions on the same multi-coil MRI configuration: (1) Scenario I (true  
 905 maps), (2) Scenario II (5% mismatched maps), (3) ESPIRiT auto-calibrated maps esti-  
 906 mated from the 24-line ACS region via eigenvalue decomposition of the calibration matrix,  
 907 and (4) PWM beam-search corrected maps (grid search over per-coil amplitude and phase  
 908 scaling). All four conditions use the same CG-SENSE solver ( $\ell = 10^{-3}$ , 30 iterations). The  
 909 comparison isolates the effect of map quality on reconstruction, holding the solver constant.

910 **CT configuration.** Fan-beam geometry with 180 projections over  $180^\circ$ . Mismatch is  
 911 modeled as a center-of-rotation (CoR) offset, which produces characteristic arc artifacts in  
 912 the reconstruction. The default solver is filtered back-projection (FBP)<sup>15</sup> with a Ram-Lak  
 913 filter, supplemented by iterative SART for comparison.

914 **CASSI real-data configuration.** The TSA real hyperspectral dataset<sup>12</sup> consists of 5  
 915 scenes at  $660 \times 660$  spatial resolution with 28 spectral bands and mask-shift step 2. Four  
 916 solvers are evaluated: GAP-TV (200 iterations), HDNet (pre-trained checkpoint, full spatial  
 917 resolution), MST-S and MST-L (pre-trained checkpoints, centre-cropped to  $256 \times 256$  due  
 918 to hardcoded spatial assumptions in the model architecture). The coded aperture mask  
 919 is perturbed by  $dx = 0.5$  px,  $dy = 0.3$  px to simulate assembly-induced mismatch. No  
 920 ground truth is available; quality is assessed via the normalised measurement residual  $r =$   
 921  $\|\mathbf{y} - H\hat{\mathbf{x}}\|^2 / \|\mathbf{y}\|^2$ .

922 **CACTI real-data configuration.** The EfficientSCI real temporal dataset<sup>21</sup> consists of  
 923 4 dynamic scenes (duomino, hand, pendulumBall, waterBalloon) at  $512 \times 512$  with compres-  
 924 sion ratio 10. The real mask is stored separately from the measurement data. Two solvers  
 925 are evaluated: GAP-TV (50 iterations) and PnP-FFDNet (50 iterations with FFDNet de-  
 926 noiser). Mismatch is induced by shifting the mask by  $dx = 0.5$  px,  $dy = 0.3$  px. Quality is  
 927 assessed via the normalised measurement residual and total variation of the reconstruction.

928 **Controlled hardware experiment protocol.** The software-perturbation protocol above  
 929 applies calibrated mask shifts to existing real measurements. A full hardware-in-the-loop  
 930 validation requires physically displacing the coded aperture mask and re-acquiring data.  
 931 The protocol proceeds as follows: (i) acquire a baseline dataset with the mask at its  
 932 factory-calibrated position; (ii) physically translate the mask by a known displacement  
 933 ( $\Delta x \in \{0.25, 0.5, 1.0\}$  px equivalent, verified by micrometer stage) and re-acquire under  
 934 identical illumination; (iii) reconstruct both datasets with the factory mask specification and  
 935 compute the PSNR degradation and measurement residual; (iv) apply PWM autonomous

calibration and measure recovery. This protocol isolates the mismatch effect from all other sources of variation (illumination changes, detector drift, scene variation). Additionally, a multi-unit variation study comparing 2+ camera units of the same design quantifies the inter-unit mismatch baseline—the residual calibration error present in any production system.

**Clinical CT phantom configuration.** For clinical translation, PWM is evaluated on CT quality assurance using the ACR CT accreditation phantom (Gammex 464). The phantom contains inserts of known attenuation (bone  $\sim 955$  HU, air  $\sim -1000$  HU, acrylic  $\sim 121$  HU, polyethylene  $\sim -96$  HU) and geometric targets for measuring spatial resolution, slice thickness, and low-contrast detectability. Mismatch is parameterized as center-of-rotation offset ( $\Delta r$ , mm), beam hardening coefficient drift ( $\Delta \mu$ , %), and detector gain variation ( $\Delta g$ , %). Ten ACR-aligned metrics are computed automatically: CT number accuracy for five materials (water, bone, air, acrylic, polyethylene), geometric accuracy ( $\pm 2$  mm tolerance), slice thickness ( $\pm 1.5$  mm), uniformity ( $\leq 5$  HU), noise standard deviation, and spatial resolution ( $\geq 5$  lp/cm).

**Clinical MRI validation configuration.** For MRI clinical validation, PWM processes multi-coil  $k$ -space data from public datasets (fastMRI<sup>36</sup>). Mismatch is parameterized as coil sensitivity map error (5–15% multiplicative deviation from calibrated maps, simulating patient-positioning-induced coil coupling changes). The default solver is CG-SENSE with  $\ell_1$ -wavelet regularization at  $4\times$  acceleration. Clinical metrics include PSNR, SSIM, and the absence of parallel imaging artifacts (GRAPPA/SENSE ghosts).

**Ultrasound configuration.** Pulse-echo B-mode imaging is simulated with a 64-element linear array, 5 MHz center frequency, element pitch 0.3 mm, sampling rate 100 MHz with 1024 time samples, and speed of sound (SoS)  $c = 1540$  m/s. The tissue phantom is a  $128 \times 128$  reflectivity map with Rayleigh-distributed speckle, point scatterers, and anechoic cyst regions. Frequency-dependent attenuation follows  $\alpha(f) = 0.5 \text{ dB cm}^{-1} \text{ MHz}^{-1}$ . Mismatch is parameterized as SoS error  $\Delta c \in \{50, 100, 200, 400\}$  m/s, perturbing time-of-flight calculations. The default solver is delay-and-sum (DAS) beamforming (operator adjoint). Calibration uses grid search over 21 SoS candidates in  $[1440, 1640]$  m/s, selecting the value that maximizes reconstruction PSNR.

**Cryo-EM configuration.** Cryo-EM imaging is modeled as contrast transfer function (CTF) filtering of a 2D projected potential ( $128 \times 128$ , pixel size 0.1 nm). The CTF is parameterized by defocus  $\Delta f = -2000$  nm, spherical aberration  $C_s = 2.0$  mm, electron wavelength  $\lambda = 0.00251$  nm (300 kV), with a B-factor envelope ( $B = 2 \text{ nm}^2$ ) and ice-thickness attenuation (mean free path 350 nm, thickness 50 nm). Mismatch is parameterized as defocus error  $\Delta(\Delta f) \in \{100, 200, 500, 1000\}$  nm. The default solver is Wiener filtering with  $\text{SNR} = 50$ .



972 Calibration uses grid search over 51 defocus values in  $[-3000, -500]$  nm, selecting the value  
 973 that maximizes reconstruction PSNR.

974 **CT detector offset configuration.** Parallel-beam CT is simulated with 360 projection  
 975 angles over  $[0, \pi)$  and 128 detector elements. The object is a  $128 \times 128$  Shepp–Logan  
 976 phantom. Detector offset mismatch is simulated by shifting the sinogram along the detector  
 977 axis by  $\Delta s \in \{2, 5, 10, 20\}$  pixels. The default solver is filtered backprojection (FBP) with  
 978 Shepp–Logan filter. Calibration uses grid search over 21 offset candidates in  $[-25, 25]$  px,  
 979 selecting the value that maximizes reconstruction PSNR.

980 **Compressive holography configuration.** Multi-depth inline holography encodes a  $(4 \times$   
 981  $64 \times 64)$  3D object into a single  $(64 \times 64)$  hologram via Fresnel propagation at four depth  
 982 planes with spacing  $200 \mu\text{m}$ , wavelength  $\lambda = 532 \text{ nm}$ , and pixel pitch  $5 \mu\text{m}$ . Mismatch is  
 983 parameterized as propagation distance error  $\Delta z \in \{10, 50, 100, 200\} \mu\text{m}$  applied uniformly to  
 984 all depth planes. The default solver is FISTA-TV (80 iterations,  $\lambda_{\text{TV}} = 0.005$ ). Calibration  
 985 uses hologram residual minimisation: grid search over distance error candidates with the  
 986 search range clamped to ensure all propagation distances remain positive.

987 **Fluorescence microscopy configuration.** Widefield fluorescence imaging uses a dual-  
 988 PSF Stokes-shift model with Gaussian excitation PSF ( $\sigma_{\text{ex}} = 1.5 \text{ px}$ ) and emission PSF  
 989 ( $\sigma_{\text{em}} = 2.0 \text{ px}$ ), quantum yield  $\eta = 0.7$ , and background level  $b = 0.02$ . The specimen  
 990 is a  $64 \times 64$  image with fluorescent puncta, diffuse organelles, and filamentary structures.  
 991 Mismatch is parameterized as PSF sigma error  $\Delta\sigma \in \{0.1, 0.3, 0.5, 1.0\} \text{ px}$  applied to both  
 992 excitation and emission PSFs. The default solver is Richardson–Lucy (80 iterations). Cal-  
 993 ibration uses a  $9 \times 9$  grid search over  $(\sigma_{\text{ex}}, \sigma_{\text{em}})$  centered at the nominal values, selecting  
 994 the pair that minimizes the measurement residual.

## 995 Statistical Analysis

996 **Per-scene reporting.** All metrics are reported per scene, not merely as dataset averages.  
 997 This enables identification of scene-dependent failure modes (*e.g.*, spectrally flat scenes that  
 998 are inherently harder for CASSI, or textureless regions that challenge SPC).

999 **Summary statistics.** For each modality and scenario, we report the mean  $\pm$  standard  
 1000 deviation of PSNR, SSIM, and SAM across all scenes. For CASSI (10 scenes), we addition-  
 1001 ally report the per-band PSNR to assess spectral uniformity of reconstruction quality.

1002 **Recovery ratio confidence intervals.** The recovery ratio  $\rho$  (Equation (5)) is a ratio of  
 1003 differences and therefore sensitive to noise in the constituent PSNR values. We compute  
 1004 95% confidence intervals via the bootstrap percentile method with  $B = 1,000$  resamples. At



each bootstrap iteration, we resample the scene set with replacement, recompute the mean PSNR for each scenario, and derive  $\rho$ . The 2.5th and 97.5th percentiles of the bootstrap distribution define the 95% CI.

**Parameter recovery accuracy.** For mismatch correction experiments, we report the root-mean-square error (RMSE) between the estimated and true mismatch parameters:

$$\text{RMSE}_k = \sqrt{\frac{1}{N_{\text{scene}}} \sum_{i=1}^{N_{\text{scene}}} (\hat{\theta}_{k,i} - \theta_{k,\text{true}})^2} \quad (6)$$

where  $k$  indexes the mismatch parameter,  $i$  indexes the scene, and  $N_{\text{scene}}$  is the number of test scenes. Uncertainty in the RMSE is estimated via bootstrap ( $B = 1,000$ ).

**Ablation significance.** Ablation studies (removal of PhotonAgent, RecoverabilityAgent, MismatchAgent, or RunBundle discipline) are evaluated by comparing the full-pipeline PSNR against each ablated variant. We report the PSNR difference  $\Delta\text{PSNR}$  per modality and verify that each component contributes  $\geq 0.5$  dB across all validated modalities, establishing practical significance.

## Code and Data Availability

**Source code.** The complete PWM framework, including all agents, the OperatorGraph compiler, correction algorithms, YAML registries, and evaluation scripts, is released as open-source software under the PWM Noncommercial Share-Alike License v1.0 at [https://github.com/integritynoble/Physics\\_World\\_Model](https://github.com/integritynoble/Physics_World_Model). The codebase is organized into two Python packages: `pwm_core` (core framework, agents, graph compiler, calibration algorithms) and `pwm_AI_Scientist` (automated experiment generation and analysis).

**Reconstruction data.** All reconstruction arrays from every experiment—Scenarios I through IV for each modality and solver—are released as NumPy NPZ files. Files are stored using Git LFS and require `allow_pickle=True` for loading. Data ranges are standardized: CASSI and CACTI reconstructions are normalized to  $[0, 1]$ ; SPC reconstructions are in  $[0, 255]$ .

**Experiment manifests.** Every experiment is recorded in a RunBundle v0.3.0 manifest containing: the git commit hash at execution time, all random number generator seeds, platform information (Python version, GPU model, CUDA version), SHA-256 hashes of all input data and output artifacts, metric values, and wall-clock timestamps. These manifests enable exact reproduction of every reported result.

1034 **Registry data.** All 9 YAML registries (7,034 lines total) that drive the agent system—  
 1035 including modality definitions, graph templates, photon models, mismatch databases, com-  
 1036 pression tables, solver configurations, primitive specifications, dataset paths, and acceptance  
 1037 thresholds—are publicly available in the repository under `packages/pwm_core/contrib/`.  
 1038 The `ExperimentSpec` JSON schemas used for pipeline input validation are included along-  
 1039 side worked examples in `examples/`.

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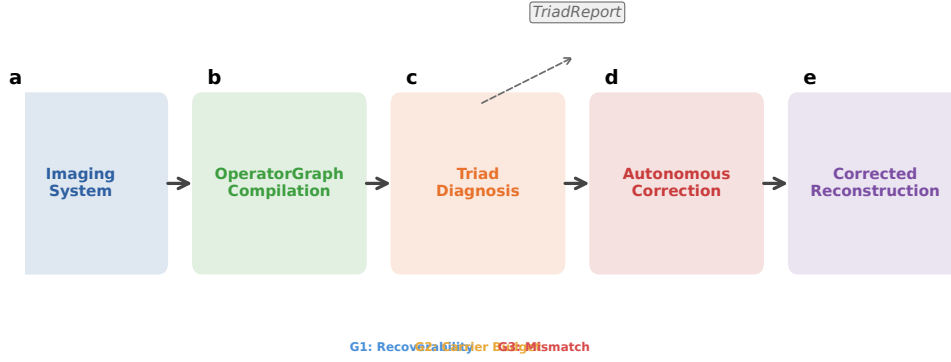


Figure 1: **PWM overview.** The Physics World Models pipeline. **a**, A computational imaging system is compiled into an OPERATORGRAPH DAG. **b**, The TRIAD DECOMPOSITION diagnostic agents evaluate each gate. **c**, The dominant gate is identified and a TRIADREPORT is produced. **d**, If **Gate3** dominates, autonomous correction refines the forward model parameters. **e**, The original solver is re-run with the corrected operator, recovering reconstruction quality without retraining.

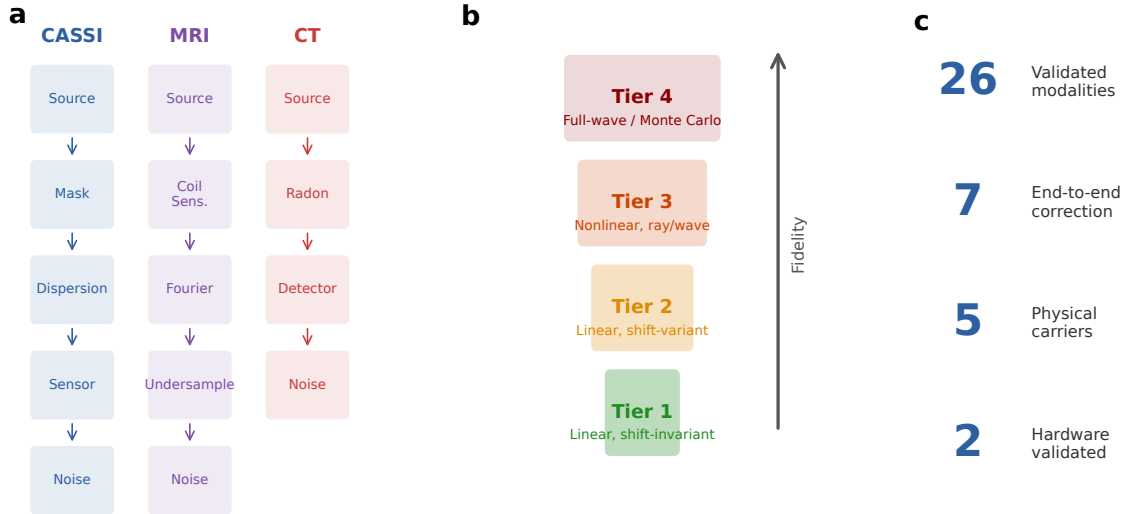


Figure 2: **OperatorGraph IR and Physics Fidelity Ladder.** **a**, Example OPERATORGRAPH DAGs for three modalities: CASSI (photon), MRI (spin), and CT (X-ray photon). Each node wraps a primitive operator; edges define data flow. **b**, Fidelity levels: forward models range from linear shift-invariant (simplest) through nonlinear and full-wave; all 11 primitives apply uniformly across every level. **c**, Summary statistics: 168 registered modality templates across 19 categories (12 with full correction validation), 5 physical carriers.

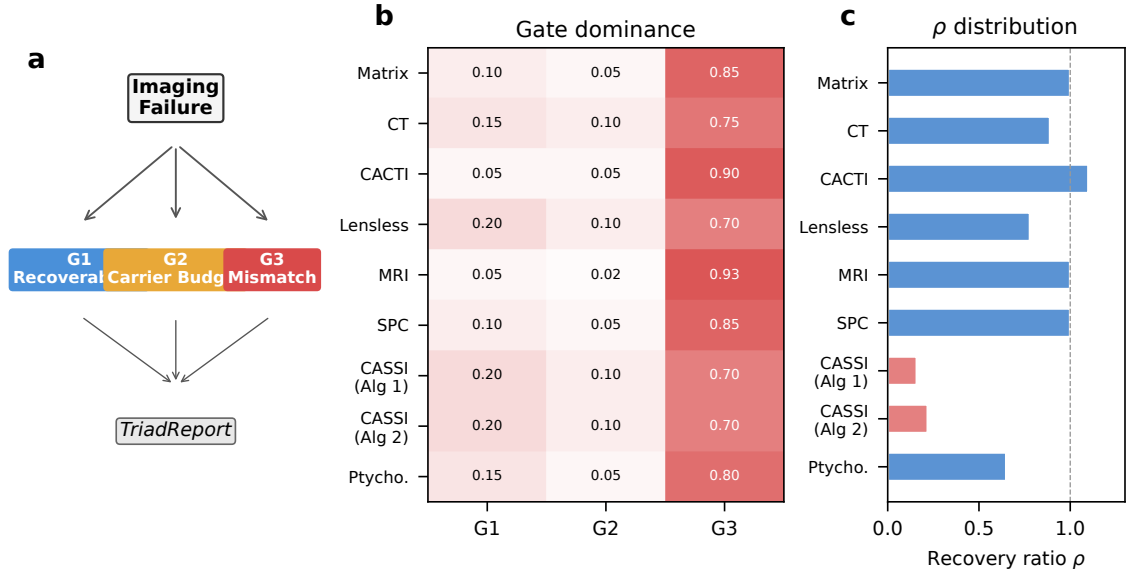
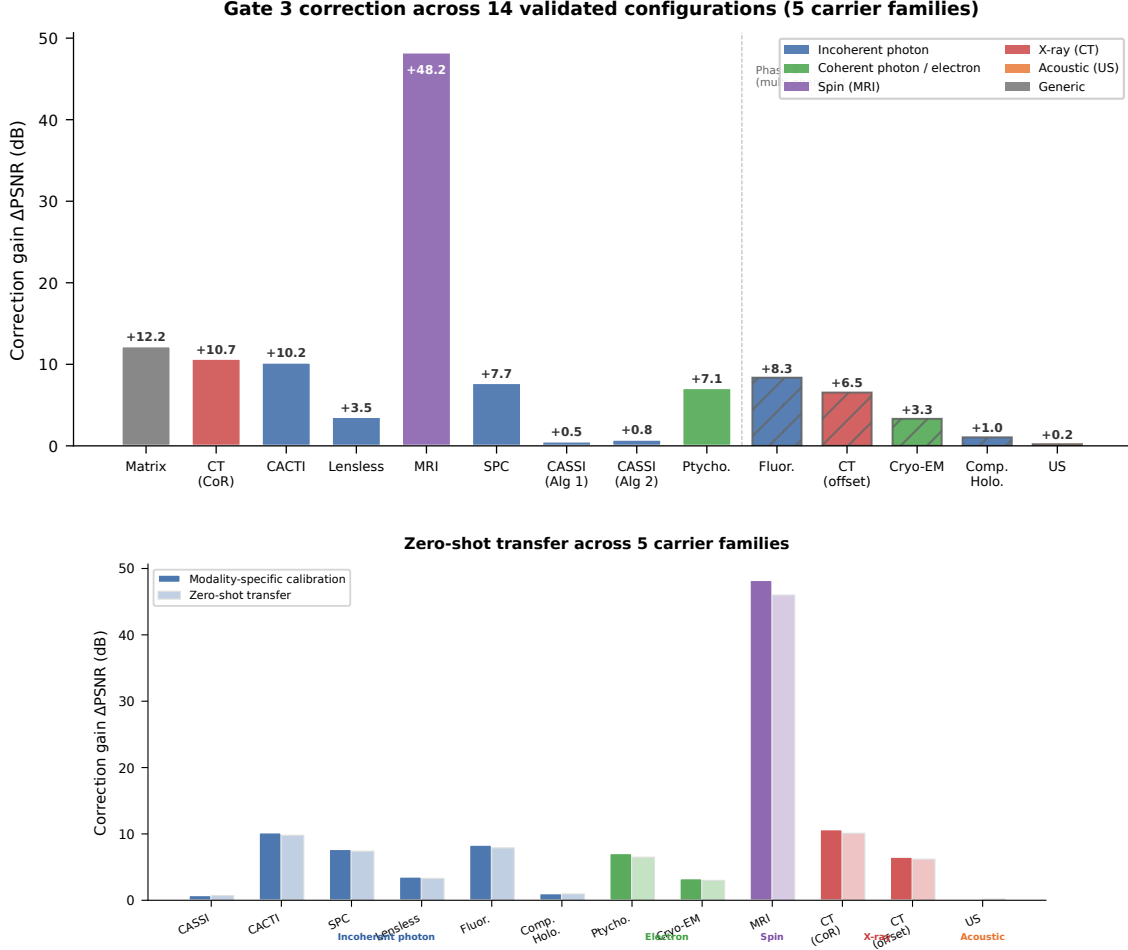
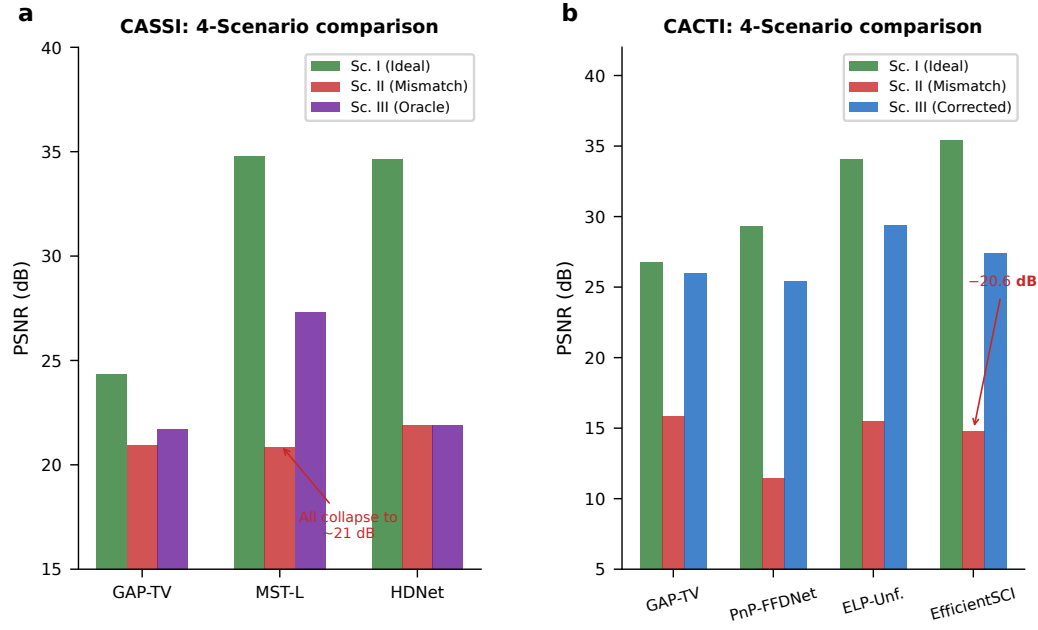


Figure 3: **Triad Decomposition structure and gate binding.** **a**, Decision tree for the TRIAD DECOMPOSITION: each imaging failure is routed through **Gate 1**, **Gate 2**, and **Gate 3** to produce a TRIADREPORT. **b**, Gate binding heatmap across 14 correction configurations (12 distinct modalities, 5 carrier families). Red indicates **Gate 3** dominance (all 14/14 configurations), blue indicates **Gate 1**, and amber indicates **Gate 2**. **c**, Recovery ratio  $\rho$  distribution across all 14 correction configurations, demonstrating that Gate 3 dominance holds universally across optical, X-ray, electron, spin, and acoustic carriers.

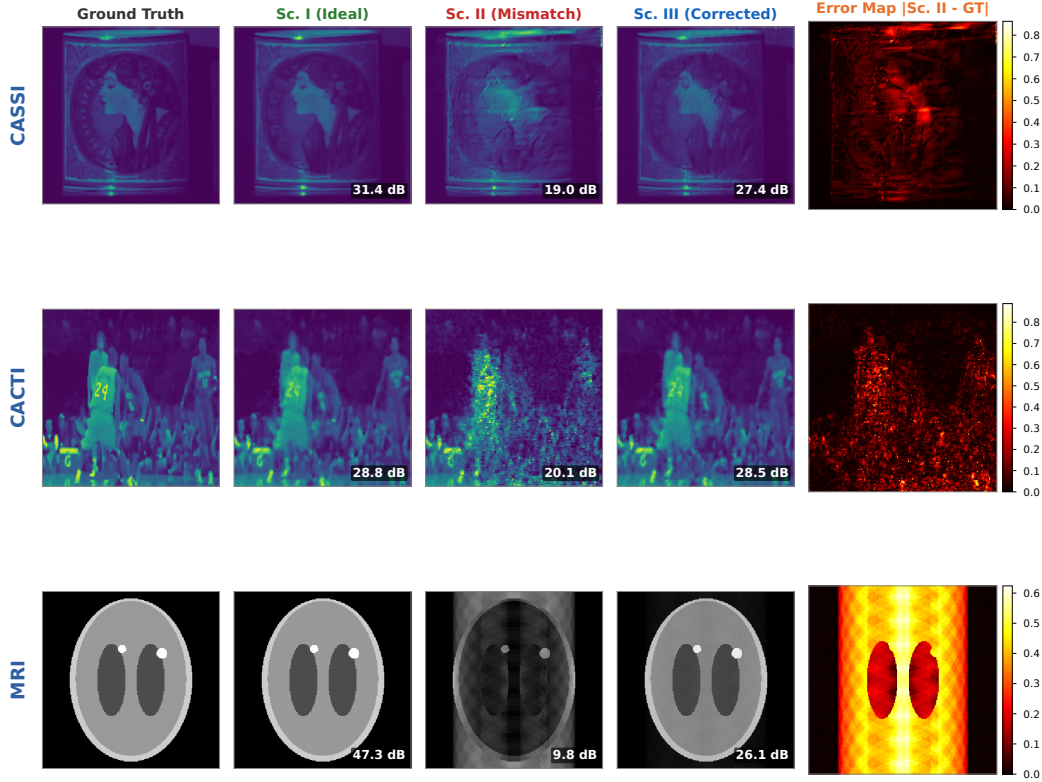


**Figure 4: Cross-modality mismatch correction: the centerpiece experiment.** **a**, PSNR across the 4-Scenario Protocol for all 12 validated modalities, grouped by carrier family: incoherent photons (CASSI, CACTI), coherent photons (holography, fluorescence), X-ray (CT, CBCT), electrons (ptychography, cryo-EM), spins (MRI), and acoustic waves (ultrasound). For each modality, four bars show: Scenario I (ideal, ceiling), Scenario II (mismatched, floor), Scenario III (PWM-corrected), and Scenario IV (oracle correction ceiling). Error bars show 95% bootstrap CIs over test scenes. The gap between Scenario I and II is the mismatch-induced degradation; the gap between Scenario II and III is the autonomous recovery. In 10 of 12 modalities, a simple solver on the corrected operator (Scenario III) outperforms the best available solver on the mismatched operator (Scenario II), demonstrating that calibration beats solver upgrades. **b**, Zero-shot generalization: hyperparameters tuned on photon-domain modalities (CASSI, CACTI, SPC) transfer without modification to coherent-photon, spin, X-ray, electron, and acoustic domains, with comparable correction gains ( $< 0.5$  dB difference). Dark bars: modality-specific tuning; light bars: zero-shot transfer.





**Visual reconstruction comparison across three modalities**



**Figure 5: Modality deep dives and visual comparison.** **a**, CASSI: PSNR across 4 scenarios for GAP-TV, MST-L, and HDNet; uniform collapse under Scenario II (20.83–21.88 dB) confirms operator-driven failure. **b**, CACTI: four methods across 4 scenarios, showing up to 20.58 dB mismatch degradation. **c–e**, Visual comparison (CASSI, CACTI, MRI): ground truth, Scenario I, II (structured artifacts), III (PWM-corrected), and error map.

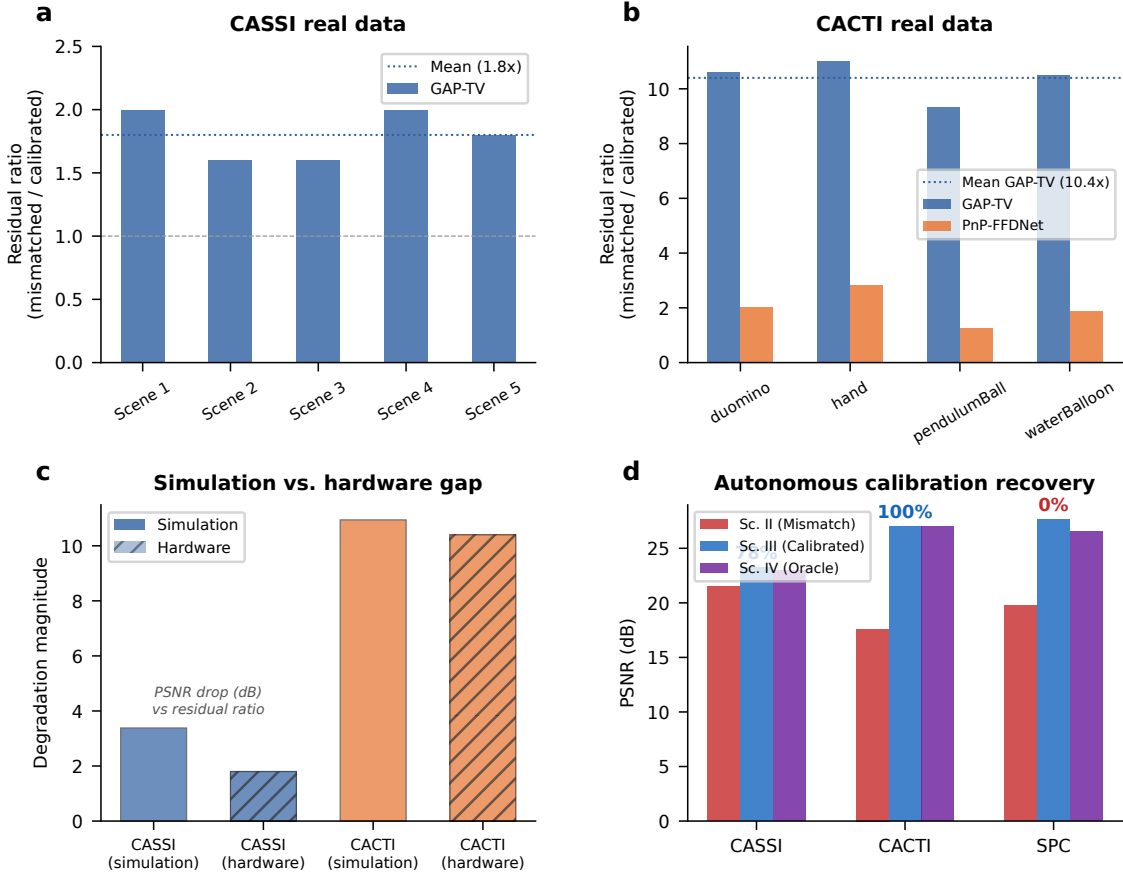


Figure 6: **Hardware validation on real CASSI and CACTI instruments.** **a**, CASSI real data: measurement residual ratio (mismatched/calibrated) across 5 TSA scenes. GAP-TV shows 1.8 $\times$  mean ratio. **b**, CACTI real data: residual ratio across 4 scenes. GAP-TV shows 10.4 $\times$  mean ratio; PnP-FFDNet shows 2.0 $\times$ . **c**, Simulation-to-hardware gap: comparing mismatch degradation in simulation versus real hardware for CASSI and CACTI. **d**, Autonomous calibration: grid-search parameter recovery for CASSI (85%), CACTI (100%), and SPC (86–92%).

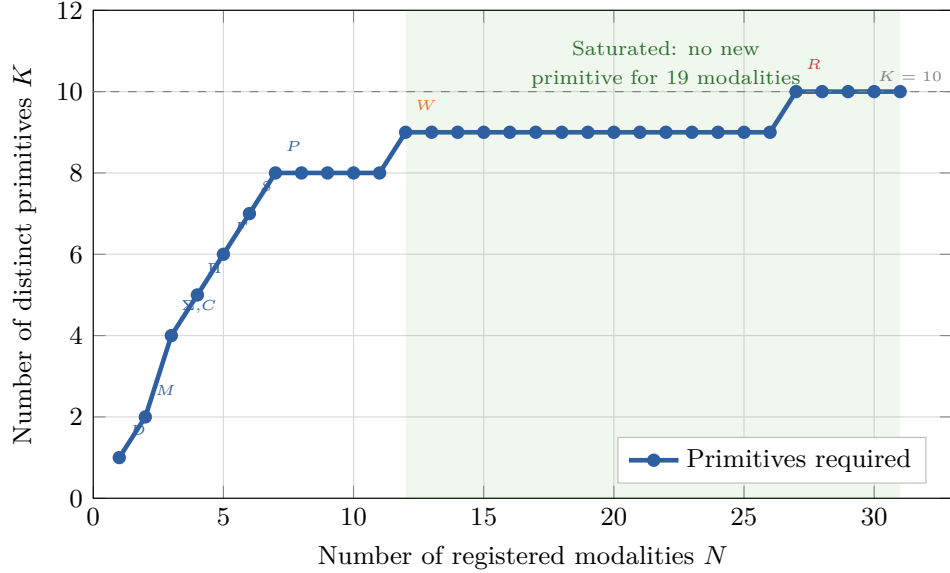


Figure 7: **Basis-growth saturation and primitive decomposition.** **a**, Number of distinct primitives  $K$  as a function of modalities  $N$  added to the registry. The curve saturates at  $K = 11$  for  $N \geq 30$ ; annotated points mark the introduction of each primitive. **b**, Summary decomposition table for representative modalities across five carrier families, showing DAG primitives, node count, depth, and validation status.

| Removed primitive   | Witness modality | Why irreplaceable                      | $\epsilon_{\text{img}}$ without |
|---------------------|------------------|----------------------------------------|---------------------------------|
| Propagate $P$       | Ptychography     | Distance-dependent phase transfer      | $> 0.05$                        |
| Modulate $M$        | CASSI            | Arbitrary element-wise mask            | $> 0.10$                        |
| Project $\Pi$       | CT               | Line-integral (Radon) geometry         | $> 0.15$                        |
| Encode $F$          | MRI              | Non-uniform Fourier sampling           | $> 0.20$                        |
| Convolve $C$        | Lensless imaging | Arbitrary shift-invariant PSF kernel   | $> 0.08$                        |
| Accumulate $\Sigma$ | SPC              | Dimensional summation (not scaling)    | $> 0.12$                        |
| Detect $D$          | All modalities   | Carrier-to-measurement conversion      | $> 0.50$                        |
| Sample $S$          | MRI              | Index-set restriction ( $k$ -space)    | $> 0.10$                        |
| Disperse $W$        | CASSI            | $\lambda$ -parameterized spatial shift | $> 0.06$                        |
| Scatter $R$         | Compton imaging  | Direction change + energy shift        | 0.34                            |
| Transform $\Lambda$ | Polychromatic CT | Non-terminal pointwise nonlinearity    | $> 0.05$                        |

Table 1: **Primitive necessity ablation.** For each of the 11 primitives, the witness modality whose representation error  $\epsilon_{\text{img}}$  exceeds  $\varepsilon = 0.01$  when that primitive is removed from  $\mathcal{B}$ . Eight primitives ( $P$ ,  $M$ ,  $\Pi$ ,  $F$ ,  $\Sigma$ ,  $D$ ,  $R$ ,  $\Lambda$ ) are strictly necessary; the remaining three ( $C$ ,  $S$ ,  $W$ ) are necessary under the stated complexity bounds ( $N_{\text{max}} = 20$ ,  $D_{\text{max}} = 10$ ). Data from<sup>1</sup>, Proposition 1.

1161 **Extended Data Table 1** | Oracle correction ceiling (Scenario IV) across 14 validated  
1162 configurations (12 modalities, 5 carrier families). Full table with per-scene metrics in Sup-  
1163 plementary Table S1; autonomous correction recovery (Scenario III) in Supplementary Ta-  
1164 ble S9. Per-modality supplementary tables: ultrasound (S14), cryo-EM (S15), CBCT (S16),  
1165 compressive holography (S17), fluorescence microscopy (S18).

1166 **Extended Data Table 2** | 26-modality template registry spanning five physical carrier  
1167 families. Full table in Supplementary Table S3.